

Techno-economic modeling and optimal sizing of autonomous hybrid microgrid renewable energy system for rural electrification sustainability using HOMER and grasshopper optimization algorithm

Timothy Oluwaseun Araoye^{a,*}, Evans Chinemezu Ashigwuike^b, Muncho Josephine Mbunwe^c, Oladipupo Idris Bakinson^d, ThankGod Izuchukwu Ozue^e

^a Department of Mechatronic Engineering, University of Nigeria, Nsukka, Nigeria

^b Department of Electrical and Electronic Engineering, University of Abuja, FCT., Nigeria

^c Department of Electrical Engineering, University of Nigeria, Nsukka, Nigeria

^d Department of Electrical Engineering, De Montfort University, Leicester, United Kingdom

^e Department of Electronic and Computer Engineering, University of Nigeria, Nsukka, Nigeria

ARTICLE INFO

Keywords:

Hybrid renewable energy
Grasshopper optimization algorithm
HOMER pro software
Modeling
And sensitivity simulation analysis

ABSTRACT

This research paper focuses on techno-economic modeling and optimal sizing of autonomous hybrid microgrid systems. The optimal configuration of the suggested stand-alone system was performed by developing a size optimization model based on the metaheuristic novel Grasshopper Optimization algorithm (GOA) method to minimize the Total Net Present Cost (TNPC), unmet load, and Cost of Energy (COE) in the Nsukka Community which comprises 88 villages. The GOA and HOMER Pro Software are employed to compare results across four possible configurations of hybrid renewable power systems (HRES). The comparative analysis between GOA and HOMER shows that configuration-4 (biogas/Diesel), emerges as the optimal solution, with a 0 % unmet load at the COE of \$0.01783 per kWh. The findings indicated that the GOA-based HRES, with a higher saturation of Biogas and photovoltaics (PV), proves to be more affordable in comparison to the HOMER-based solutions. The reduction in the COE and NPC of renewable energy for peak demands highlights the growing importance of biogas generators as an affordable local power supply to meet energy demands. This underscores the potential of the GOA in optimizing hybrid renewable energy systems for remote communities, producing an economically viable and sustainable energy solution.

1. Introduction

Renewable energy sources like wind, hydropower, and biogas can help create a more sustainable and eco-friendly economy by reducing the use of nonrenewable energy sources. However, using renewable resources in a flexible, stand-alone framework comes with several limitations. The hybrid renewable energy such as grid, wind, and solar power sources are combined with other sources to solve these problems. Thus, the efficiency of the power produced can be determined by the availability of renewable resources generated [1,2]. Rural electrification is hindered by many factors such as limited grid transmission, rugged topography, and widely dispersed valleys with a sparse population that is characterized by lower income and high load density. When HOMER software is used to design a hybrid power system, the various expenses

associated with the process, such as the energy cost and net present cost, must be determined. Genetic algorithms (GA) were used by the author [2] to develop the hybrid system for meeting the energy requirements of different demand segments within the communities of Chamarianagar in the Indian state of Karnataka's southern region. Author [3] proposed a method of sizing hybrid renewable energy sources (HRES) for generation of a steady power supply. A comparative analysis of hybrid energy systems based on demand-side monitoring (DSM) and load shifting was developed by the author [4]. The demand-side control plan was recommended as the best course of action in the absence of the demand-side monitoring strategy. According to Ref. [5], various hybrid energy system configurations were analyzed in six different geographical zones of Nigeria. HOMER software was applied to analyze the economic feasibility with sensitivity scenarios between \$1.1 to \$1.3 based on COE and NPC. Author [6] applied statistical method to

* Corresponding author.

E-mail address: timothy.araoe@unn.edu.ng (T.O. Araoye).

<https://doi.org/10.1016/j.renene.2024.120712>

Received 27 January 2024; Received in revised form 9 May 2024; Accepted 24 May 2024

Available online 25 May 2024

0960-1481/© 2024 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

Nomenclature

DSM	Demand-side monitoring	GSA	Gravitational Search Algorithm
COE	Cost of Energy	BEES	Battery Energy Storage System
NPC	Net Present Cost	CAES	Energy-compressed Storage
GA	Genetic algorithms	GOA	Grasshopper Optimization Algorithm
HRES	Hybrid Renewable Energy Sources	SOC	State of Battery Charge
WCA	Water Cycle Algorithm	A_{pvg}	PV generator Surface Area (m^2)
MFO	Moth-flame optimization	η_{pvg}	Conversion Efficient
PHES	Pumped Hydro Energy Storage	η_r	Reference Efficiency for Solar Modules
EMS	Energy Management System	β	Temperature Index for Silicon Photovoltaic Cells
PV	Photovoltaic	T_{clef}	Standard Temperature for Solar Cells ($0^\circ C$)
FPA	Flower Pollination Algorithm	G_{ir}	sun radiation on a global scale(W/m^2)
HES	Hybrid Energy System	T_c	temperature of the photovoltaic cell
RES	Renewable Energy System	T_a	ambient temperature in $^\circ C$
BBO	Biogeography-based optimization	NOCT	standard operating cell temperature in $^\circ C$
PSO	Particles Swarm Optimization	V_{mppt}	voltage at the point of peak power
ABC	Artificial Bee Colony	α	temperature index
HS	Harmony Search	G_{it}	irradiation estimation value (in kW/m^2)
ELBO	Learning Educating-based Optimization	G_{st}	standard radiation value (in units of $1 kW/m^2$)
BFPSO	Butterfly –Particle Swarm Optimization	T_a	temperature's variable (K)
GWOPSO	Grey Wolf Optimization-particle Swarm Optimization	I_{ph}	symbolizes the photocurrent
NN	Neural Networks	I_{rs}	saturation current
AI	Artificial Intelligence	q	electrons charging
DE	Differential Evolution	N_s	total variable of series cells
ACOA	Ant Colony Optimization Algorithms	K	Boltzmann's Variable constant
CCS	Cycle Charging Strategy	A_i	optimum ratio of diodes
MILP	Mixed-Integer Linear Computation	E_{PV}	total PV energy produced
NLP	Nonlinear Optimization	$E_{Em}(t)$	amount of non-available renewable energy
MCGP	Multi-Choice Goal Programming	$\sigma CHGb$	Battery charging efficiency
MOEAs	Multi-Objective Evolution Algorithmic	αCC	Battery charging regulator
LPM	Linear Learning Model	SOC_{min}	SOC minimum variable
CS	Cuckoo Search	SOC_{max}	SOC maxi ydropower
SA	Simulated Annealing	γ_{hyd}	hydro turbine reliability, which is 65 %
BFO	Bacterial Foraging Techniques (BFO)	ρ_{hyd}	density of water ($1000 kg/m^3$)
ACA	Ant Colony Algorithm	Q_{model}	flow rate model
ABC	Artificially Bee Colonies Algorithm	h	accessible head
HAS	Harmony Search Method	g	gravitational acceleration ($9.81 m/s^2$)
TAC	Annual Total Cost	E_{BOG}	energy output of the biogas gasifier
LCOE	Lowest Cost of Energy	η_{BG}	system conversion efficiency
LPSP	Loss of power supply probability	BV_{BG}	biogas digester calorific value ($4700 kcal/kg$)
HPP	Hybrid Power Plants	RC	replacement cost
FL	Federated Learning	FC	fuel Cost
LPM	longest prefix match	O&MC	total operation and maintenance cost
EDE	Enhanced differential evolution	CC	total investment cost
TLBO	Teaching learning based optimization	K_{WT}^{max}	peak variable of wind turbines
SSA	Salp Swarm Algorithm	K_{PV}^{max}	peak variable of PV modules
AC	Aho-corasick Algorithm	K_{Bat}^{max}	peak variable of batteries
GWO	Grey Wolf Optimization	ub_d and lb_d	the upper and lower limit along the d-th axis
		$\hat{T}_d$	optimal solution in d-th dimension space

determine the potential of solar and wind energy in Nigerian rural areas using data from available meteorological sources. The author used HOMER software to design and size the best possible system of hybrid power components in terms of both economics and technology. Author [7] also compared the two optimal system combinations: Diesel-PV-battery and Wind-PV-diesel-battery systems with the typical system. The second configuration (Wind-PV-diesel-battery) generates less economic feasibility of COE and NPC with 0.409 \$/kWh and \$69, 811 respectively.

Author [8] proposed a bi-level system that makes use of decision models and a multi-objective optimization method to generate sustainable electricity in remote area. Author [9] developed WCA and MFO algorithms to power a transmitter for a radio signals station in India, the

techno-economic optimal configuration of a PV-Battery-Biogas-PHES driven hybrid power system was done along with GA. Based on three techniques, including simulated annealing, harmonic search, and chaotic search, the author [10] presented an innovative hybrid optimization methodology for the proper sizing of a hybrid autonomous power system. The feasibility study was conducted to determine the reliability index of the system. Author [11] performed an economic analysis feasibility of a fuel tank-PV off-grid hybrid power plant to provide electricity in rural regions of Egypt. With the Flower Pollination Algorithm (FPA), the author was able to determine the entire yearly cost, system efficiency, and Probability of power failure.

The authors of Refs. [12–14] observed that off-grid electrification system have tremendous impact on economic and environmental

performance in rural areas. HES are less expensive, according to the analysis done in specified rural electrification. The extra electricity produced by the hybrid system may be sold back to the grid [15]. By optimizing a grid-tied HES through changing peak load while maintaining constant energy demand, and vice versa, COE can be reduced with a drop in peak demand [16]. Energy-storing technology would be needed in the standalone HES to store the excess generated electricity by RES. The author in Ref. [17] produced a detailed sensitivity model for the system and discussed the effects of various methods of storage on the performance of the hybrid microgrid system. According to the author's [18] prediction, pumped storage is a more dependable and economical option for storing electrical energy. The comparison of various techniques is discussed by the author [19] as a means of enhancing the network's power quality and voltage profile. Authors [20,21] discussed various methods for predicting electricity costs. The methods for reducing harmonic distortion and enhancing the power quality of the network were anticipated by the author in Ref. [22]. Author [23] discussed several parameters to be considered when modeling a photovoltaic (PV) system in HES and the best method for performing the realistic values of the solar PV array modeling parameters. Author [24] applied various ab/dq frames rather than a single abc/dq frame. The author's findings provided a better phase balancing method for hybrid energy systems with grid inverter interfaces. In the case of an imbalanced current, the proposed method can produce independent references for every phase. The author [25] applied BBO and PSO algorithms to demonstrate the load-switching approach for HRES. The load-shifting strategy, according to the authors, could lower the COE, and also discovered that the BBO method outperforms PSO in terms of performance. The Author in Ref. [26] provided the environmental pollution assessment and techno-economic design for an off-grid-connected HES of a Nigerian rural town. The author proposed a novel voting-based energy management for a small rural community's HES. The new suggested strategy was shown to be dependable, economical, and environmentally friendly. Author [27] applied cost-effective technology to determine the performance index of pumped system storage. To determine the best electricity for Morocco, the Author [28] determined external variables of electrical power, and economic analysis. The research thoroughly examined the viability of isolated HRES for PV-wind plants for 24 cities. The author [29] compared GA, PSO, and HOMER methods for optimizing the hybrid system's size. The author observed that PSO are found to be more economical than Genetic Algorithm (GA) and HOMER. The research conducted by Author [30] also considered three energy management techniques which are maximum shaving load, cycle charging, and load following for an ideal hybrid System model. In comparison to the other two options, the cycle charging approach produced more inexpensive outcomes. Author [31] employed the artificial bee colony (ABC) method to determine the ideal hybrid system component sizing to meet the electrical demand consumption of a remote Indian village. The obtained results were shown to be more accurate when compared to HOMER and PSO. The author [32] applied a Genetic Algorithm (GA) to maximize the size of the integrated-grid hybridized Wind/SPV system and electrification of remote areas in Spain. The sensitivity evaluation was performed to determine the accuracy of the system. It was shown that the computed NPC of green energy systems negatively correlated with the efficiency of the systems but shows high positive correlation with cost. The size optimization of an integrated-grid photovoltaic power system plant for a group of six households in Iran was described by the author [33] using the harmony search (HS) technique. The findings showed that installing a PV system could be cost-effective if current electricity rates continue to rise. The PSO-based approach was utilized by Author [34] to optimize the design of integrated-grid hybrid Wind/PV Energy systems to minimize COE and maximize reliability. The cost savings of the optimized hybrid system were compared to those of the real utility grid. The author [35] proposed the use of various optimization techniques, including PSO, GA, butterfly PSO, and learning educating-based optimization

(ELBO), to minimize the multi-objective function while sizing an autonomous HRES to generate electricity to the isolated region of Barwani district, India. Economic, social, and technical factors were all considered to form a multi-objective function. The best combinations are chosen from a variety of options. The findings showed that ELBO outperforms BFPSTO, PSO, and GA. The regenerative osmosis desalination machine was powered by an ideal grid-tied Wind/PV hybrid system that was developed by the author [36] using a hybrid GWO-PSO (grey wolf optimization-particle swarm optimization) technique. To minimize costs and greenhouse gas emissions, two distinct configurations were compared which include Wind/PV/battery and Wind/PV/hydrogen system. The configuration of PV/battery/wind proved to be more economical and ecologically beneficial, according to the outcomes. Additionally, the hybrid PSO–GWO technique is determined to be superior when compared to PSO and GWO alone. It was also discovered that adding DG can lower the cost. Additionally, a sensitivity evaluation was carried out to examine the effect of differences in wind speed and solar radiation on cost, and it was discovered that solar radiation had a greater effect compared to wind speed. The literature has shown that intelligent strategies perform better than simulation software in terms of potential outcomes. Furthermore, off-grid hybrid system optimization has accounted for the majority of work. Also, HOMER simulation software has been used for the majority of the work on the grid-connected scenario. Researchers have been heavily investigating several intelligence strategies inspired by natural phenomena in recent years for optimization. These techniques include neural networks (NN), genetic algorithms (GA), particle swarm optimization (PSO), differential evolution (DE), ant colony optimization algorithms (ACOA), harmony search (HS) etc.

1.1. Cycle charging strategy (CCS)

Cycle charging, which involves using renewable power generators to meet the load demand and charging batteries through the excess energy when the level of charge is within the specified limit state of charge (SoC) ranges according to Fig. 1 [37]. The batteries will start operation if demand load surpasses the energy produced by renewable generators. The diesel generator will immediately begin meeting the net consumption and charging the batteries in response to a subsequent increase in demand. This kind of scenario occurs when batteries and renewable generators cannot meet the demand load for 1 h. The diesel power plant in CCS is design to meet the net load and instantaneously charge the batteries. Three cases are presented in this paper to illustrate the method as shown in Fig. 1.

1.1.1. Case 1

This situation occurs when renewable energy sources can both meet consumption and charge batteries independently. Hence, this cause the use of S3 and S4 switches.

1.1.2. Case 2

This occurs when the demand marginally exceeds the overall generation from renewable sources which trigger the operation of battery

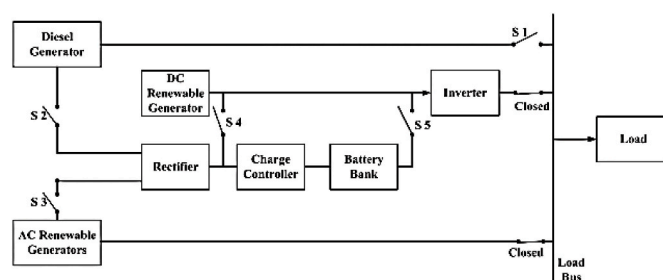


Fig. 1. Working of cycle charging strategy [29].

storage. The Case 3 is used instead of this one unless SOC_{min} is lesser than battery State of charge (SoC). Here, the switching mechanism S5 is in use.

1.1.3. Case 3

If the demand rises further, a diesel-powered generator is also incorporated to satisfy the unmet net demand and provide extra energy for battery charging if the battery's maximum state of charge is not reached. All five switches which include S_1 , S_2 , S_3 , S_4 , and S_5 are operational in this state.

1.2. Optimization techniques

Energy system optimization aims to produce the best outcomes under certain conditions and boundaries while considering the objectives of stakeholders. Optimization involves maximizing the best use of all available resources which includes energy sources, sizes, budgets, controls, and energy management while maintaining the constraints and rules of the grid. Optimization can be attained by choosing several variables to minimize or maximize (as shown in Fig. 2).

The three types of common optimization approaches include artificial technique, classical, and hybrid as shown in Fig. 3, and are utilized in a variety of applications [38–45]. The primary objectives of optimizations may include minimizing energy loss, maximizing land use efficiency, reducing redundant infrastructure, optimizing existing facilities by integrating various energy sources into a single electrical plant, and achieving improved profitability [46–49].

Classical optimization techniques deal with complex variables due to the extremely use of mathematical models that discover universally the best possible outcomes in a deterministic approach. These includes graphical, probabilistic, analytical, and iterative analyses and generate energy models using the differential calculus method. These techniques are constrained, nevertheless, in cases when the objective functions are not differentiable or continuous. Classical approaches include iterative techniques like mixed-integer linear computation (MILP) [50–54], nonlinear optimization (NLP) [55], multi-choice goal programming [50–54], multi-objective evolution algorithmic approaches (MOEAs)

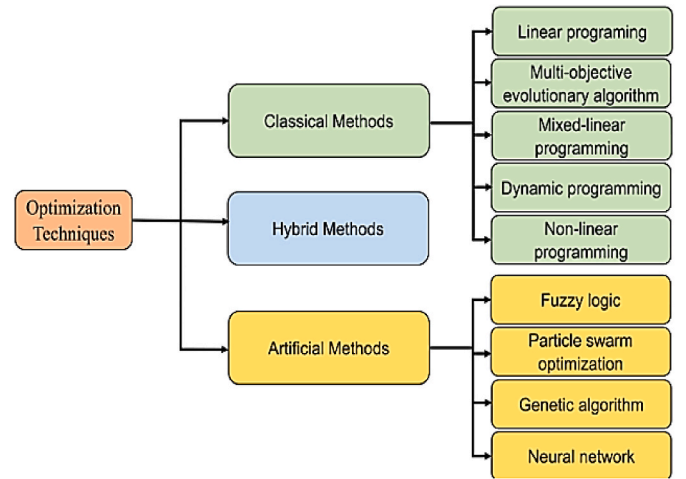


Fig. 3. Microgrid Optimization Methods [67].

[51], and linear learning model (LPM) [56]. These techniques reduce energy shortfalls, lower system expenses, and maintain a steady electricity supply.

Different factors, like geography and demand patterns, can affect a hybrid system's optimal configuration. On the other hand, the constrained capacity of classical optimization techniques limits the researchers' ability to utilize them effectively. For flexible designs, predictive methods provide statistical justifications, whereas deterministic approaches employ resources and load demand as established variables with predictable time-series fluctuations. Applying outlines and optimization functions on the same plane while concentrating on the implementation zone is the goal of graphic creation techniques. The literature [38–46] describes the use of artificial methods for sizing renewable sources, including cuckoo search (CS), simulated annealing (SA), bacterial foraging techniques (BFO), ant colony algorithm (ACA), artificially generated bee colonies algorithm (ABC), harmony search

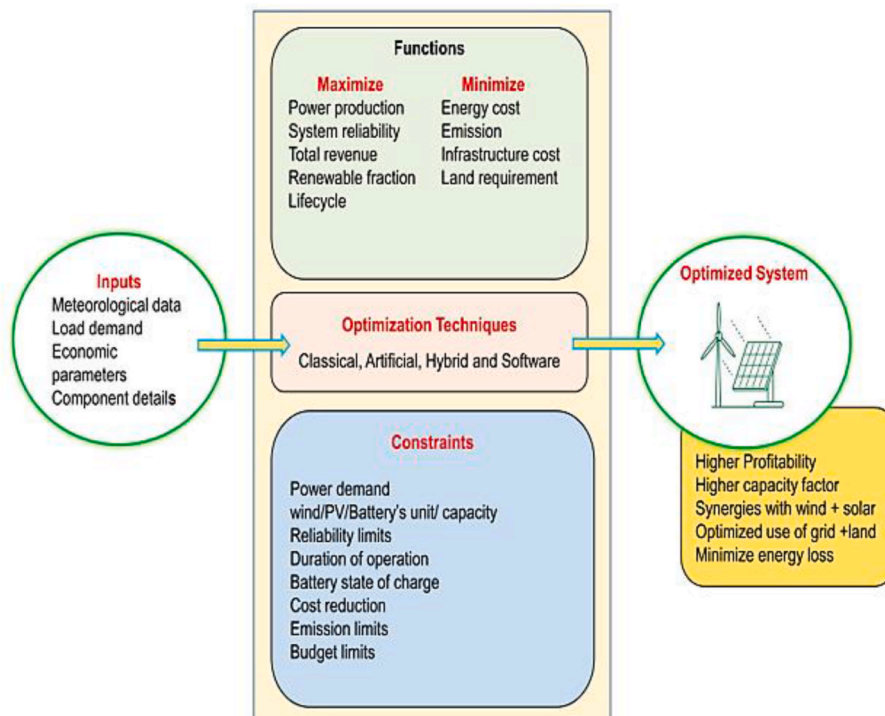


Fig. 2. Microgrid Optimization process [67].

method (HSA), and GA. These algorithms successfully handle the non-linear properties of energy production system components and the sporadic fluctuation of wind and solar energy sources. In contrast with classical methods, the algorithms proved tremendous advantages such as shorter calculation times, higher accuracy, and faster convergence rates. However, this section primarily focuses on approaches that researchers have recently and frequently used. GA is used by evolutionary heuristic search techniques in renewable energy hybrid structures to optimize parameters. This method has been successfully used by researchers in several contexts, including the scheduling of day-ahead plans for hybrid electricity plants [57], the optimization of hybrid wind-solar-storage plant design and layout [58], and the balancing of life cycle expenses, system energy utilization, and electrical loss risk in wind-PV-battery hybrid configurations [59]. However, the effectiveness of these methods depends on constant wind speeds and steady PV cell voltage output, which can be difficult in actual renewable energy systems. The goal of the PSO optimization algorithm is to minimize LCOE while maintaining a suitable production range. It can be applied to lower energy storage costs [60] in hybrid power plants (HPPs), lower energy costs (COE) [61], lower LPSP, lower annual total cost (TAC), and lower emissions [62], determine multi-objective optimization challenges to guarantee system reliability, and lower unmet load, total cost, and emission levels of fuel [63].

1.3. Summary and evaluation

Artificial intelligence techniques enable hybrid systems to scale more quickly and easily than traditional methods. AI approaches are capable of solving complicated problems by using multi-objective functions. Though recursive procedures in iterative techniques can assist reduce constraints, but unintentionally may neglect important factors. Though the complicated programming of the algorithms poses difficulties, artificial intelligence provides flexibility and good results for challenging tasks. The various techniques used in hybrid systems are discussed in Table 1 and Table 2.

1.4. HRES energy storage sources

To improve reliability, a Battery Energy Storage System (BESS) can be integrated with the Hybrid Renewable Energy System (HRES). When there is increase in energy use or a dip in renewable energy production because of weather conditions, the Battery Energy Storage System (BESS) accumulates and releases electrical energy generated by renewable sources like wind turbines or solar PV. Essentially, the HRES's consistent and environmentally friendly electricity supply is largely dependent on the BESS. Table 3 presents a summary of the most often used BESSs in the literature to date, along with a breakdown of their benefits and drawbacks. It also makes it easier to find articles depending on the technology of choice by identifying each technology utilized in the reviewed articles.

1.5. Novelty of the study

The study explored autonomous microgrid renewable energy hybrid models in four different modes of configuration in rural areas of Nsukka Communities. A novel model has been developed in this paper, drawing on the existing literature as well as the research gaps. Therefore, the research applied HOMER and Grasshopper Optimization Algorithm for proper sizing and techno-economic problems in the villages. The research applied four HRES combinations to find the best configurations with the lowest possible NPC and COE values. In addition, the optimized system has an appealing payback period, lower emissions, and is economically effective and environmentally friendly. Moreover, sensitivity analysis is provided for changes in the annual wind speed, flow rate, and biomass output together with the energy cost and net preset cost.

Table 1

A comparison of different methods with their respective advantages and disadvantages [64–67].

Method/ Software	Advantages	Disadvantage	Reference
Iterative	- Simple to operate and effective in detecting threats at an early stage. - Straightforward to implement in programming. - Reducing expenses can be accomplished very effectively by using linear programming methods or linear variable parameters.	Linear adjustments in decision variables lead to suboptimal solutions as the operation fail to optimize factors such as the slope angle of PV modules and the installation height of wind turbines, which have a more substantial impact on costs.	Chauhan [68]
Probabilistic	- Because of its stochastic nature, the system can arbitrarily choose the best mode of action given the available information. - Time-series information is not necessary for basic scaling procedures.	- Optimization focuses on a restricted set of performance parameters, which may not be ideal for identifying the best solutions. - Less effective in capturing the dynamic fluctuations in performance within a hybrid system.	Ganguly [69]; Lian et al. [70]
GA	GAs are robust instruments capable of handling numerous parameters, discrete variables additional integers, and non-differentiable attributes, concurrently. The parallel processing capability enables the simultaneous assessment of multiple methods, thereby increasing the probability of attaining peak iterations. GA has demonstrated superiority in environmental and cost analysis.	Because GAs are designed to conduct local searches, the operation takes a long time but requires fewer computational resources than the analytical approach.	Iweh et al. [71]; Riaz et al. [72]
PSO	Ability to tackle complicated problems with ease through rapid selection of the most effective universal solution and parallel analysis of data.	In complicated problems with various parameters, the system performs poorly in local searches, which frequently leads to early convergence.	Dubey et al. [73]; Gad et al. [74]; Wang et al. [75]; Gupta and Srivastava [76]
ACO	The algorithm demonstrates analogy, flexibility, and the capacity to quickly identify the best options through feedback processes. Additionally, it can be combined with a different algorithm	When dealing with finite problems, improper parameter selection can lead to rapid convergence, decline, and failure to arrive at the best possible outcomes. It can also pose challenges when	Gupta and Srivastava [76]

(continued on next page)

Table 1 (continued)

Method/ Software	Advantages	Disadvantage	Reference
CS	to create a dependable and strong solution. In comparison with particle swarm optimization, it requires fewer iterations to reach the global minimum. The efficiency of the algorithm is enhanced by using Levi's flying attribute, which promotes convergence near globally the best possible outcomes. It has effective optimization capabilities, resilient unpredictable search routes, and a high degree of amenability to algorithmic hybridization.	dealing with continuous problems. • A faulty starting variable could cause a sluggish convergence rate by getting caught in a regional search.	Shen et al. [77]
SA	• Programming is simple, robust, and adaptable, allowing for seamless switching between global and local search patterns. The process of reaching the best outcome is predictable and systematic.	• The system has poor performance and takes a long time to compute.	Iweh et al. [71]
HS	• The approach works without demand derivative information, requires fewer configurable control variables, and does not require an initial configuration of a decision variable.	• Shows inconsistent and delayed convergence, with little capacity to limit and modify search parameters.	Dubey et al. [73]
GWO	• Less parameters. • Simplified implementation.	• A poor convergence speed in subsequent iterations has lowered the algorithm's accuracy and precision and prevented local search presence.	Wang et al. [75]
iHOGA	• The capacity to employ a control technique directly when sizing HRES.	• Achieving specific objectives and solving non-linear problem scenarios are the main priorities.	Saiprasad et al. [78]
RETScreen	• Significantly reduces costs associated with finding and assessing possible energy developments when compared to Homer Pro.	• The task entails manually gathering information and organizing it into an Excel sheet or workbook.	Ramli et al. [79]
Coyote optimization	The method exhibits rapid convergence and	In particular, when dealing with	[80]

Table 1 (continued)

Method/ Software	Advantages	Disadvantage	Reference
algorithm (COA)	operates effectively with a wide range of optimization problems, especially multimodal and nonlinear instances.	situations that include several local optima, the COA may have problems finding exact responses and may become stuck at the local maxima. This is what happens when an algorithm gets stuck on a less-than-ideal solution rather than finding the best option overall.	
AWSA (Amended Water Strider Algorithm)	Modifying the AWSA to accommodate different types of optimization problems is achievable.	A poor initial solution can make it difficult for the algorithm to find the best response, which can have a significant impact on how effective AWSA performs.	[81]
HOMER	The use of visual data representation can simplify the process of implementing complex systems by facilitating in the recognition of the system optimal configuration.	Analysis of reliability cannot be performed effectively.	[82–84]

The following are the paper's primary contributions:

- A distinctive hybrid renewable energy technology (HRET) was designed for cost and size optimization in rural areas.
- The paper applied an EMS concept that provides efficient energy management in an autonomous community with optimal sizing model to determine the lowest-cost architecture of the HRES using GOA and HOMER Pro software
- The paper considered four configurations of autonomous HRES based on available resources observed in areas with the lowest possible values of NPC and COE using multi-objective mathematical constraint programming.
- Also, the research observed that accessibility and availability of animal manures in the study villages contributes to the efficiency of the Biogas generator system.

2. Methodology

A collection of rural, off-grid villages without electricity is chosen using the present load demand. An hourly load in the rainy and harmattan seasons, as well as the possible evaluation of several locally accessible sources, are used to create a load profile based on the data gathered for the overall population and resources survey. The required sources, utilizing the cycle charging approach already stated, can satisfy this sequence of hourly change in the load profile. The initial stage involves completing the demand and hourly resource evaluation. Renewable generators are offered precedence in meeting demand, and any additional power is utilized to charge the batteries. If the battery's SoC is not reached, then the diesel generator will start operating to meet any additional demand hence, the renewable generators and batteries will take over. The battery is charged using the extra energy that the renewable energy produces while operating in two modes to meet the net demand. The HES's energy cost is optimized using the GOA technique once the process is finished.

Table 2

A summary of different research works focuses on a single objective's optimization techniques.

Optimization Method	System Configurations	Optimization Function	Constraints	Reference
FL and GA	PV/Wind/Battery	Total cost minimization	Battery State of Charge and Energy management	Abdelhak et al. [85]
Integer LPM, ACA	PV/Wind	Total cost Minimization.	Wind turbines, Number of PV panels, and batteries	Fetanat et al. [86]
SA and PSO	Battery/PV/Wind	Minimization of the present total cost.	Hybrid components parameters, Battery SoC, energy not used.	Ahmadi et al. [87]
CS	Battery/PV/Wind	Minimization of the present total cost.	Load Seasonal variation	Sanajaoba and Fernandez [88], Peng et al. [89]
CS, SA, Improved HS	Wind/PV/Battery/reverse osmosis	Minimization of the total LCC	Wind turbine, PV panel surface area, LPSP, number of batteries, and SOC	Khan et al. [90]
EDE, TLBO, and SSA	PV/Wind	Reliability and Minimization of TAC	Hybrid system parameters, DOD, and LPSP	Khan et al. [91]
TLBO and Jaya	Battery/PV/Wind	Minimization of TAC	Hybrid system parameters, SoC, LPSP	Javed et al. [92]
Firefly, AC, GA, PSO algorithm	Battery/PV/Wind	NPC	Hybrid system parameters, SOC	
PSO and Crow	Battery/PV/Wind	Minimization of COE	Power supply distribution and load planning	Guneser et al. [93]
Sine cosine and GWO Algorithm	PV/Wind/H ₂	Minimization of LCC	Hybrid system parameters quantity	Jahannooosh et al. [94]
GA and PSO	Battery/PV/Wind	LPSP, energy probability loss and LCC Minimization.	Wind turbine, batteries, PV numbers	Ma et al. [95]
Nelder Mead and PSO Algorithm	Wind/PV/fuel cell	Power loss Minimization	Bus voltage, Energy management	Senthil et al. [96]
GSA and PSO	Wind/PV	Minimization of energy total loss, Voltage profit Maximization.	Load constraints, m bus voltage Power flow, capacity of PV/wind	Radosavljevic et al. [97]
PSO and Biogeography optimization	PV/Wind/battery	Minimization of index system reliability and cost.	Power management between load demand and supply.	Aliabadi et al. [98]

2.1. Area selection

Nsukka is a town in Enugu State, Nigeria, with a total land area of 3402 km², and located between longitudes 6° 54' and 7° 54' east and latitudes 6° 30' and 7° 54' north. The region consists of 88 villages and the urban center of Nsukka, which is known as a university town due to the establishment of the University of Nigeria. Thirty-five (35) of the eighty-eight (88) rural communities were chosen at random for further investigation. In particular, a count of the rural infrastructure facilities in the 35 chosen localities was done. Seven lakes are combined to form the Opi Lake Complex, also known as the Opi Lake System, which is situated in the Nigerian village of Nsukka, Enugu State. Each of these seven lakes connects to the others to form a waterway. Ozzi, Uhre, Ogelube, Orufu, Iyi, Ikpa Eze-agu, and Nkpo make up the lake complex. Ozzi Lake is situated in Ugwu-ozioke and is 7 km broad and 9 m deep, which allows a canoe to travel from one end to the other. The lakes of Ogelube and Eze-agu Iyi are relatively close to one another, separated by only around 50 m of land. Between Ogelube and Eze-agu Iyi lakes, the Uhre River flows, offering a sandy beach, crocodiles, and other aquatic life. Four non-electrified communities in the Nsukka region were chosen for the study. There are 1875 people living in the 512 households that make up the study. The absence of an electrical grid in the community makes it challenging to supply energy to these locations. There is a plethora of renewable energy sources available in this research area, including hydro, animal manure, wind, and solar energy, all of which are used autonomously.

2.2. Energy demand estimation

The load data is gathered from the villages via surveys containing a range of sections, such as home load, agricultural demand, community demand, and business load sections, according to the requirements for energy within the scope of the study. The health center, elementary school, stores, street lamps, University, and minor industrial loads made up the majority of the energy consumption. The estimated daily total demand for energy in kWh and peak load are; 60 kWh and 9 Kw for configuration-1, 2355 kWh and 313.4Kw for configuration-2, 974 kWh and 104.10Kw for configuration-3 and 745 kWh and 93.21Kw for configuration-4 which is shown in Figs. 4–7.

2.3. Source estimation

Renewable energy sources which includes solar, hydro and wind have a lot of potentials in this area. Throughout these latitudes and longitudes, the average wind speed, daily solar radiation, and ideal system sizing were calculated using the HOMER Pro software. Fig. 8 shows the Average Monthly electricity production. Fig. 9 shows Monthly Average Variation of wind speed.

2.3.1. Annual solar radiation

Notably, the clearness index varies from 0.241 in October to 0.539 in May, and the monthly average variance in solar radiation ranges between 2.5 kWh/m²/day and 5.3 kWh/m²/day in October and March respectively according to Fig. 10. The annual solar radiation and average clearness index are 4.0 and 0.4 kWh/m²/day, respectively. Both the harmattan (December to March) and rainy (April to October) seasons have been utilized for estimating the demand load.

2.3.2. Scaled annual wind speed

The research location's annual average wind speed, as depicted in Fig. 7, is 3.79 m/s on average each year.

2.4. Modelling of system components

Modeling the components of a hybrid energy system is an important first step toward size optimization since it provides information on the system's performance in various scenarios. The following explains the mathematical modeling of the suggested HRES sections:

2.4.1. Mathematical modeling of photovoltaic system

The PV arrays are composed of interconnected solar cells that produce the required output current and voltage ratings [111]. Due to low costs of maintenance and operations and, the absence of both carbon emissions and gasoline costs, photovoltaic systems have become among the most frequently recognized source of power. The PV technology's power generation can be calculated using equation (1):

$$P_{PV} = \eta_{pvg} X A_{pvg} X G_{ir} \quad (1)$$

With A_{pvg} (m²) as the PV generator Surface Area, η_{pvg} is the conversion

Table 3
Types of BESS and their benefits and drawbacks [99,100].

BESS	Advantage	Disadvantage	Reference
Energy-compressed storage (CAES)	Cost-effective, minimal self-discharge, and extended service lifespan.	The significant upfront investment requires considerable space, and installation is restricted by geographical limitations	[101]
Battery Ni-Cd	Minimal maintenance needs, substantial energy density, and superior reliability.	This product comes with elevated expenses and is susceptible to a phenomenon referred to as “battery memory.”	[102]
Battery lead-acid	This product offers a moderate energy density, requires a low initial investment, and is readily accessible. Moreover, it operates without the need for a cell management system.	This product is characterized by a short life cycle, low efficiency, ventilation prerequisites, and the necessity for appropriate disposal of depleted batteries	[103, 104]
Battery Li-ion	This product is notable for its extended life cycle, high energy density, high efficiency, and comparatively small size. Furthermore, it is situated in a region that sees swift advancements in technology.	Due to the specific design needed and the potential hazards of battery core rupture, this product has high initial capital expenses.	[105]
Hydrogen-based (HESS)	This product offers a broad power range and is virtually free of contaminants.	Because of the hydrogen storage tank, this device has low efficiency, slow response times, high costs, and installation limitations.	[106]
Supercapacitor (SESS)	It has a long lifespan, a large power capacity, and great efficiency.	It is relatively expensive and has a poor energy density.	[107]
Flywheel (FESS)	It has an extensive life cycle, a high power capacity, and quick charging capabilities.	This device has several drawbacks, such as significant self-discharge, safety concerns, and expensive cost because it requires a separate vacuum chamber.	[108]
Gel batteries	Because of its outstanding recharge behavior and long service life, this product is a superb fit for applications with strong cycle requirements.	The fact that gel batteries have a lesser current capacity than other battery technologies is one of its drawbacks. Compared to some other battery technologies, the battery may be costlier.	[109, 110]

efficient, given by (2):

$$\eta_{pv} = \eta_r X [1 - \beta X (T_c - T_{clef})] \quad (2)$$

Where η_r is the reference efficiency for solar modules, β is the temperature index for silicon photovoltaic cells which remain constant, T_{clef} is the standard temperature for solar cells (0°C), G_{ir} is the sun radiation on a global scale (W/m^2), T_c is the temperature of the photovoltaic cell, T_a is

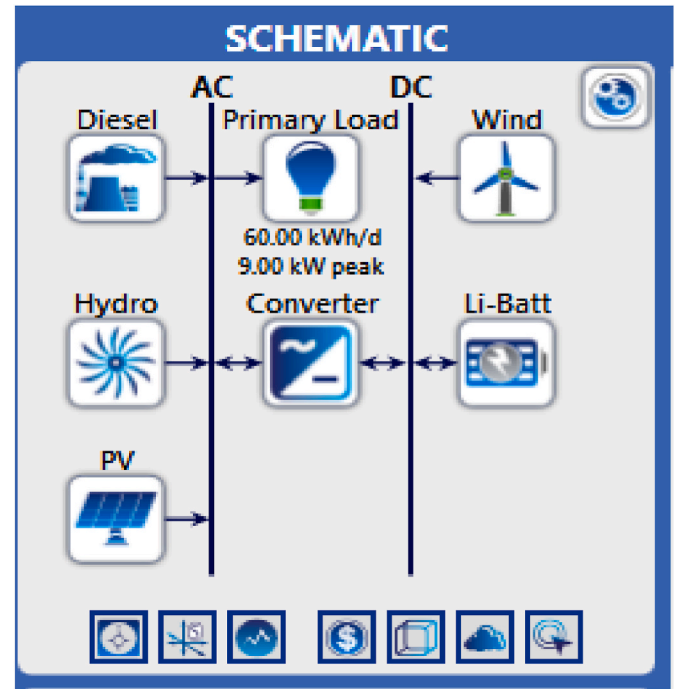


Fig. 4. Schematic of Configuration_1.

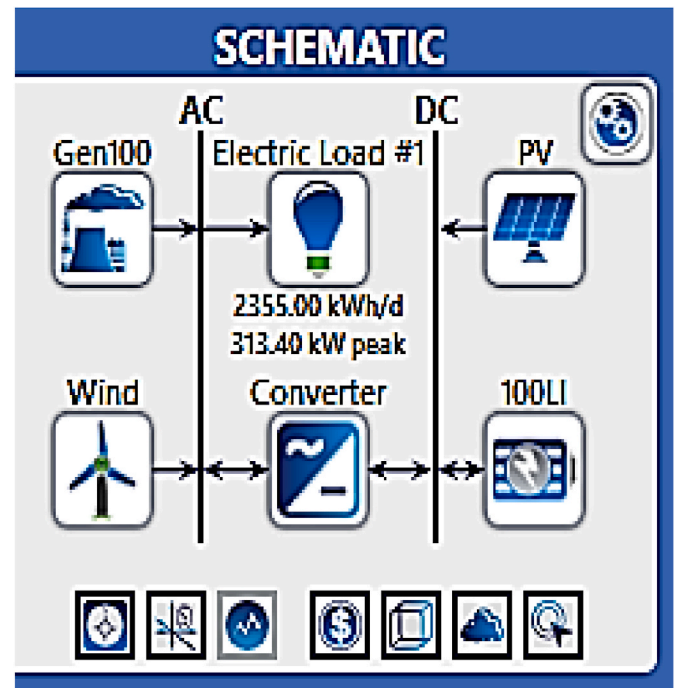


Fig. 5. Schematic of Configuration_2.

the ambient temperature in $^\circ\text{C}$, NOCT is the standard operating cell temperature in $^\circ\text{C}$.

$$T_c = T_a + \left(\frac{\text{NOCT} - 20}{800} \right) X G_{ir} \quad (3)$$

$$V_{pv} = V_{mptt} \left[1 + 0.0539 \log \left(\frac{G_u(t)}{G_{st}} \right) \right] + \alpha (T_a(t)) + 0.02 G_u(t) \quad (4)$$

where V_{mptt} displayed the voltage at the point of peak power, α is the

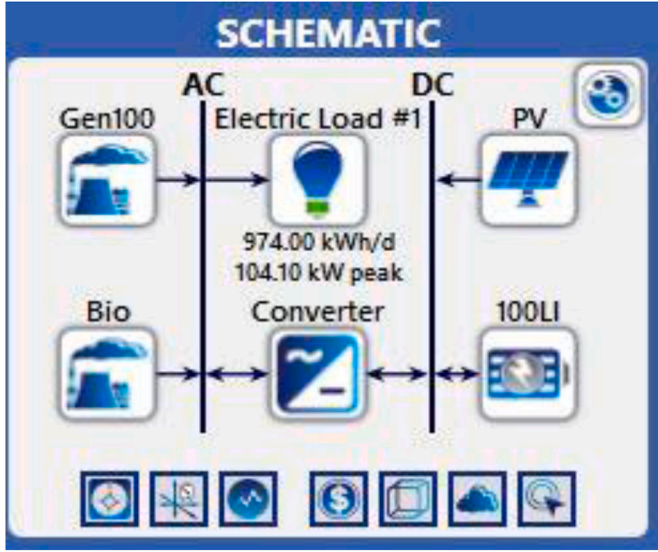


Fig. 6. Schematic of the Configuration-3.

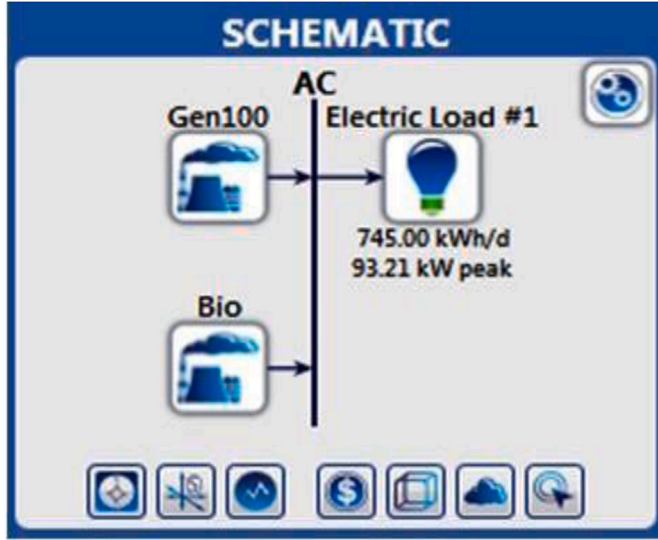


Fig. 7. Schematic of the Configuration-4.

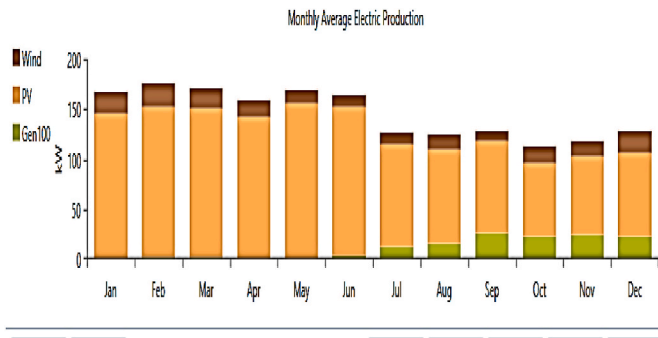


Fig. 8. Average Monthly electricity production.

temperature index, G_{tt} is the irradiation estimation value (in kW/m^2), G_{st} is the standard radiation value (in units of 1 kW/m^2), and T_a stands for the temperature's variable (K).

A solar PV mechanism's current output (I_{PV}) was calculated using

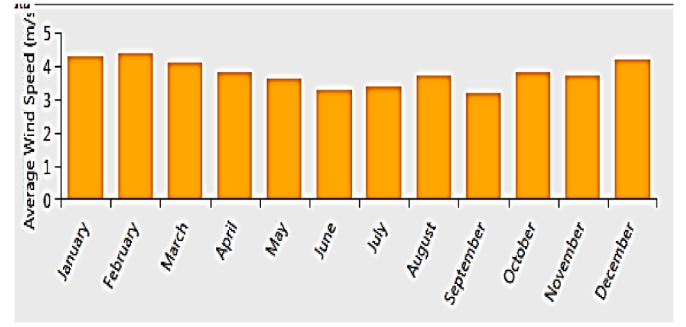


Fig. 9. Monthly Average Variation of wind speed.

Equation (5):

$$I_{PV} = I_{ph}(t) - I_{rs}(t) \left[\exp \left(\frac{qV_{PV}}{N_s K T_a(t) A_i} \right) - 1 \right] \quad (5)$$

where I_{ph} symbolizes the photocurrent, I_{rs} represents the saturation current, q represents the electrons charging, N_s represents the total variable of series cells, K is Boltzmann's Variable constant, and A_i is the optimum ratio of diodes.

The total energy produced by solar photovoltaic (PV) panels (E_{PV}) was provided by Equation (6).

$$E_{PV}(t) = \frac{(N_{PV} \times V_{PV}(t) \times I_{PV} \times \Delta t)}{1000} \quad (6)$$

where N_{PV} is the quantity of PV modules and Δt is the 15-min (15×60) second step time.

2.4.2. Mathematical modeling of battery bank

The connected batteries and the state of each charge at a given moment are used to examine the overall amount of electricity consumed and produced. The quantity of batteries available at a given time is expressed by the equation below [112]:

$$E_{Batt}(t) = E_B(t-1) + E_{Em}(t) \times \alpha_{CC} \times \sigma_{CHG} \quad (7)$$

Where $E_{Em}(t)$ is the amount of renewable energy that is yet available and battery charging efficiency is represented by σ_{CHG} , whereas the system's charging regulator is denoted by α_{CC} .

The quality of battery charging is indicated in the equation below:

$$SOC_{min} \leq SOC(t) \leq SOC_{max} \quad (8)$$

Where SOC_{min} is the SOC minimum variable, SOC is the state of battery charge, while SOC_{max} is the SOC maximum variable.

The minimum SOC variable is:

$$SOC_{min} = 1 - DOD \quad (9)$$

DOD (%) is the depth of the discharge batteries.

2.4.3. Wind turbine system

The wind power turbine profile was applied to develop a mathematical model of the wind turbine system. According to author [113], the fitting curve method controls the wind turbine models. Equations (10) and (11) are used to calculate the power generated by wind turbines (PWT):

$$\left. \begin{aligned} P(t)_{WT} &= 0 && \text{for } m \leq m_i \text{ and } m > m_0 \\ P(t)_{WT} &= P_1 M_2 + R_1 M + K_1 && \text{for } m_i < m < m_1 \\ P(t)_{WT} &= P_2 M_2 + R_2 M + K_1 && \text{for } m_1 < m < m_2 \\ P(t)_{WT} &= P_3 M_2 + R_3 M + K_3 && \text{for } m_2 < m < m_0 \end{aligned} \right\} \quad (10)$$

where M represents the wind turbine's mechanical speed, m_i denotes its break-in speed, m_0 is its break-out speed, and P , R , and K are the



Fig. 10. Monthly Average variation of solar irradiance data.

characteristic equations' coefficients. The energy generated by the system of wind turbines (E_{WT}) can be stated using equation (11).

$$E_{WT} = \frac{K_{WT} \times P(t)_{WT} \times \Delta t}{1000} \quad (11)$$

Where K_{WT} represents the number of wind turbines.

2.4.4. Modeling of the synchronous diesel generator

The synchronous diesel generator (DG) based on an hourly basis is analyzed [114] using equation (12):

$$D_f(t) = \alpha_D P_{Dg}(t) + \beta_D P_{Dgr} \quad (12)$$

Where $D_f(t)$ is the synchronous generator's hourly fuel consumption in L/h, P_{Dg} represents the average power per hour of the generator in kW, P_{Dgr} is the rated power output of the synchronous diesel generator in kW, α_D and β_D are the fuel consumption curve coefficients, L/kWh.

2.4.5. Modeling of hydropower

The electrical power of hydro turbine can be determined by equation (13) [115].

$$P_{hyd} = \frac{\gamma_{hyd} \cdot \rho_{hyd} \cdot g \cdot h \cdot Q_{model}}{1000} \quad (13)$$

P_{hyd} is the estimated hydropower, γ_{hyd} is the hydro turbine reliability, which is regarded as 65 %, ρ_{hyd} is the density of water (1000 kg/m³), Q_{model} is the flow rate model, h is the accessible head and g is the gravitational acceleration which is 9.81 m/s².

2.4.6. Biogas electrical system

The electrical energy produced by the biogas generator was calculated using the following equation, which was contingent on the availability of cattle dung [112]:

$$E_{BOG} = \frac{\text{Available Biogas} \left(\frac{m^3}{day} \right) \times BV_{BG} \times \eta_{BG} \times \Delta t}{860 \times h_{BG}} \quad (14)$$

where E_{BOG} is the energy output of the biogas gasifier; η_{BG} is system conversion efficiency, and BV_{BG} is the biogas digester calorific value (4700 kcal/kg).

3. Problem formulation

The target function and constraints that make up the problem formulation are the same:

3.1. Objective function

The net present cost includes several expenses, including fuel, capital, maintenance, replacement, and operating costs. The objective function, which is provided by equation (15) [116], is to calculate the net present cost using the suggested system's evolutionary algorithm.

$$\text{Total Net Present Cost (TNPC)} = CC + O\&MC + RC + FC \quad (15)$$

where RC is the replacement cost, FC is the fuel Cost, O&MC is the total operation and maintenance cost, and CC is the total investment cost. Equation (16) provided the overall hybrid component of the total capital cost [116]:

$$CC = \{ (\beta_{PV} K_{PV} + \beta_{WT} K_{WT} + \beta_{WT} K_{WT} + \beta_{Hy} K_{Hy} + \beta_{GEN} K_{GEN} + \beta_{Bat} K_{Bat} + \beta_{Inv} K_{Inv}) \} \quad (16)$$

The following equation (17) was used to determine the overall maintenance and operation costs for each component of the hybrid system [116]:

$$O\&MC = \{ (\varphi_{PV} K_{PV} + \varphi_{WT} K_{WT} + \varphi_{WT} K_{WT} + \varphi_{Hy} K_{Hy} + \varphi_{GEN} K_{GEN} + \varphi_{Bat} K_{Bat} + \varphi_{Inv} K_{Inv}) \} \times \sum_{j=1}^1 \left(\frac{1+u}{1+i} \right)^j \quad (17)$$

$$FC = \sum_{j=1}^1 \left(\frac{1+u}{1+i} \right)^j (\gamma_A (FF_{diesel})) \quad (18)$$

$$COE = \frac{TNPC \times CRF}{\sum_{t=1}^{8760} E_{gen}(t)} \quad (19)$$

Equation (19) is used to calculate the cost of energy (COE) and overall net present cost in the research region for the renewable energy hybrid system analysis [117]:

$$\text{Capital Recovery Factor (CRF)} = \frac{\tau(\tau+1)^\sigma}{\tau(1+\tau) - 1} \quad (20)$$

The capital recovery factor, represented by supplied equation (20) as follows [118], is dependent upon the rate of yearly interest (τ) and the diversity of plants (σ).

$$E_{bat-min} \leq E(t)_{bat} \leq E_{bat-max} \quad (21)$$

3.2. Constraints

The optimal objective function is formulated as follows using the following constraints:

3.2.1. Battery storage constraint

A battery bank's storage capacity at any given hour "t" falls between its lowest and peak capacities. The restriction is then stated as [112].

3.2.2. Bounds constant

According to author [112], the restriction of lowest and upper boundaries for wind, solar, and battery storage systems is as follows:

$$K_{WT} = \text{Integer} \quad 0 \leq K_{WT} \leq K_{WT}^{\max} \quad (22)$$

$$K_{PV} = \text{Integer} \quad 0 \leq K_{PV} \leq K_{PV}^{\max} \quad (23)$$

$$K_{Bat} = \text{Integer} \quad 0 \leq K_{Bat} \leq K_{Bat}^{\max} \quad (24)$$

where $K_{WT}^{\max}$ is the peak variable of wind turbines, $K_{PV}^{\max}$ is the peak variable of PV modules and $K_{Bat}^{\max}$ is the peak variable of batteries.

3.2.3. Power reliability constrain

The addition of unmet demand constraints is contingent upon power reliability. As stated by Ref. [112]:

$$\text{Unmet load} = \frac{\text{Yearly load Supplied}}{\text{Total load}} \quad (25)$$

The overall system variable cost for each component is analyzed with different parameters in the mathematical modeling of HRES configuration. Authors [112,117,119,120] provide descriptions of the many parameters considered in the system components.

4. Grasshopper optimization (Goa) and HOMER software

4.1. Application of HOMER pro software for optimization

In these experiments, the hybrid renewable energy system is modeled using HOMER program (developed by the National Renewable Energy Laboratory, USA) (HOMER program). Design and simulation includes optimal sizing conditions of HRES. HOMER is a program designed to develop a sophisticated model operation for grid-integrated and hybrid power plant planning.

4.1.1. Simulation analysis

The designer's selection of components essentially determines the functionality of the simulation system. HOMER performs system's functionality which increases both size and component productivity. The hybrid power system in this study was evaluated, considering solar PV, hydro, biogas, wind turbine, diesel, converter, and battery. Based on the electricity demand, the simulation assessment selects the optimal dynamic planning and system design. In addition, HOMER calculates the whole cost of the hybrid system, including fuel, maintenance and repair, capital, and other costs.

4.1.2. Sensitivity analysis

Running a sensitivity study on the influence of numerous factors on the overall cost and viability of a RE-dependent microgrid system is highly recommendable. Generally, evaluation is carried out by modifying the input variable and observing the output's behavior change. There are numerous types of technical and economic input in HOMER that can be changed due to environmental conditions. The cost of wind, hydro, and solar turbines, as well as the cost of batteries and generator fuel, are the sensitivity factors. The list of several HRES components will be taken into consideration throughout development, ranging from a smaller to a more significant TNPC. Batteries (Batt), converters (Conv.), diesel generators (DG), hydro generators (HG), wind turbine generators (WT), and PV systems make up the proposed HRES. Diesel, hydro, and wind generators are connected to the power generated by AC buses, while solar PV systems are connected to DC buses for power generation. Batteries are used to store excess energy, which is subsequently

transmitted to the coast via ready-made power cables. In the absence of extra energy generation, power transfer is applied.

The generated excess energy after charging the batteries can be fed into the utility grid when the HRES is connected with it, resulting in almost little excess electricity. In these systems, peak-shaving reduced excess electricity generated by the HRE from power grid purchases. To dump the excess electricity, however, a dummy load must be taken into consideration if selling to the grid is not feasible. The potential to prevent too much electricity is now a crucial survey feature in HRES design. One possible way to prevent excess energy is to build an additional flexible energy system or adopt techniques that make better use of surplus electricity, such as the energy export strategy in larger energy sources or water storage objectives in smaller form.

4.2. Grasshopper optimization algorithm

According to Saremi et al. [121], a rule that mimics the natural hunting and swarming behaviors of grasshoppers is anticipated to operate an innovative and smart swarm algorithm. Grasshoppers, a common insect, can sustain dangerous injuries that impact crop productivity and agriculture [121,122]. The two life cycles are nymph and adult. The adult phase's characteristic is determined by sudden wide-range movements, while the nymph phase's characteristic is determined by sluggish, little steps [121]. The nymph and adult movements caused the grasshopper improvement algorithm's diversification and intensification phase. Below is a mathematical representation of grasshopper swarm behavior:

$$M_i = K_i + W_i + G_i \quad (26)$$

where M_i represents the grasshopper's i-th location, G_i represents the i-th force of gravitational pull, K_i its social interaction, and W_i its change in wind temperature. Often, grasshopper random behavior can be produced by rewriting equation (1) as follows.:

$$M_i = n_1 K_i + n_2 G_i + n_3 W_i \quad (27)$$

where the numbers that are generated at random in the boundary [0, 1] comprise n_1 , n_2 , and n_3 . The social interaction of grasshoppers (K_i) can be written as follows:

$$K_i = \sum_{j=1, j \neq i}^N K(d_{ij}) \hat{d}_{ij} \quad (28)$$

where N is the overall quantity of grasshoppers, also $d_{ij} = |K_j - K_i|$ displays the Euclidean grasshopper distances for the i and j th., $\hat{d}_{ij} = \frac{K_j - K_i}{d_{ij}}$ indicates the i -th dispersion of the grasshopper module vector j -th variance, and W stands for the intended social forces:

$$S(r) = Y \exp^{-\frac{r}{u}} - \exp^{-r} \quad (29)$$

The symbols u and Y represent the distance of the attraction velocity and the degree of attraction, respectively. Grasshopper social iteration is the term commonly used to describe attraction and repulsion. As a result, the gap line fluctuates between [0,15]. Within the range of [2.079,4], the duration of attraction will gradually grow and then diminish. The repulsion interval occurs at the boundary [0, 2.079]. At 2.079, the distance between two grasshoppers in succession is zero, meaning that there is no force involved. The grasshopper's iteration that is suggestive of the comfort region is depicted in Fig. 11 below. The gravity force, G_i , is represented by the equation below.

$$G_i = h \hat{e}_q \quad (30)$$

where $\hat{e}_q$ shows the unit vector's centroid of the earth, and h stands for the gravitational constant.

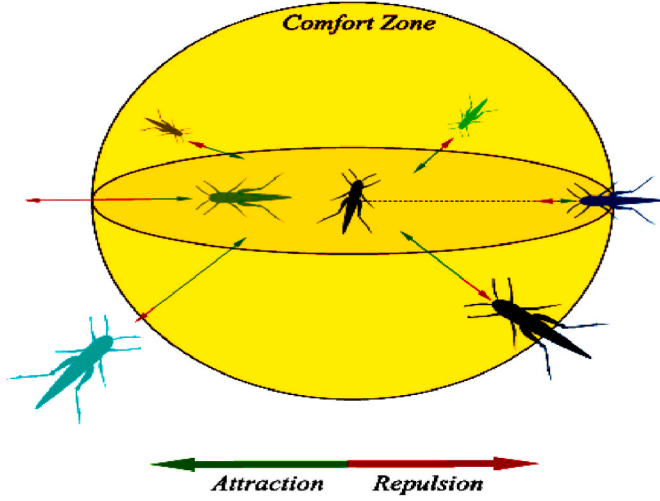


Fig. 11. The comfort zone of Grasshopper's Communication [121].

The definition of the diffusion wind A_i is:

$$W_i = R\hat{e}_s \quad (31)$$

Where $\hat{e}_s$ signifies a wind direction segment vector, while R stands for the drift constant.

Consequently, the following formula can be obtained by substituting the values of K , W , and G :

$$K_i = \sum_{j=1}^N W(K_j - K_i) \frac{K_j - K_i}{d_{ij}} - h\hat{e}_q + R\hat{e}_s \quad (32)$$

Because the grasshopper can travel quickly enough to reach the comfort zone and cause the system swarm to diverge from the target position, equation (32) cannot be employed directly to address optimization problems [121,123]. The model modification is updated below:

$$K_i^d = V \left[\sum_{j=1}^N V \frac{ub_d - lb_d}{2} W(K_j^d - K_i^d) \frac{K_j - K_i}{d_{ij}} \right] + \hat{T}_d \quad (33)$$

Where ub_d and lb_d represent, respectively, the upper and lower limit along the d -th axis., $\hat{T}_d$ indicates the optimal solution in d -th dimension space. As a result, in formula 1, the K variable and the K component are connected, and W typically bends toward the optimal solution. G and $\hat{T}_d$ are equal to 0. Particle swarm optimization's inertia weight ω and parameter V_1 's inertial weight are probably the same, and A_0 represents loudness (BA) or E_a in GWO, respectively [124]. This is applied to reduce the grasshopper's speed as it approaches the tangential point (food). It provides a fair balance between the tangent zone's intensification and diversification as a result. By using V_2 , the attraction zone, the grasshoppers' comfort zone relative to the total number of iterations, and the repulsion zone can all be reduced. Below is the equation that expresses the single variables V_1 and V_2 [121].

$$V = V_{\max} - t \frac{V_{\max} - V_{\min}}{t_{\max}} \quad (34)$$

The variable $t_{\max}$ defines the maximum number of iterations, whereas t indicates the current iteration. V and $V_{\max}$ indicate the lowest and highest values of V , respectively. Consequently, the position of the grasshopper, the optimal worldwide, and the grasshopper's location within the swarm determine the main position. It is prevented from

becoming liberated in municipal optima by this consequence [121,125].

4.3. The grasshopper optimization pseudo-code

- 1 Conduct an arbitrary baseline population P_i for the Grasshoppers ($i = 1, 2, \dots, n$).
- 2 Establish the variation of the most iteration $t_{\max}$, $V_{\max}$, and $V_{\min}$.
- 3 Verify each grasshopper's fitness level (P_1)
- 4 T = the most efficient respond
- 5 If ($t < t_{\max}$), however, do
- 6 Adjust the victimization equation (34) for V_1 and V_2
- 7 when the population of all grasshoppers, represented by the parameter N , is equal to $i = 1$ to N do
- 8 Regulate the gap variability of the grasshopper [1,4].
- 9 Update equation (33) to reflect the current victimization of grasshoppers.
- 10 Choose the current exterior limits at Grasshopper feedback.
- 11 finalizing for
- 12 Update the variable T to a higher resolution.
- 13 evaluate $t = t + 1$.
- 14 pause wherefore
- 15 examine the most efficient trajectory action for T

In the suggested system, the variables wind, PV, hydro, diesel, biogas, and battery are taken into consideration. The GOA toolbox is utilized to optimize problems to determine the fitness function. The findings indicate that the functions that are stated are taken into consideration in a minimal amount of generations.

5. Results and discussions

As discussed, the primary aim of the research is to determine the optimal size and cost of an island microgrid hybrid power plant to meet the demand load of the area. HOMER Pro and GOA are utilized to simulate the suggested combination based on the provided database.

5.1. HOMER simulation results

Several combinations of cost and size variables are developed following hourly simulation using HOMER. Four configurations out of numerous possibilities are suggested as illustrated in Table 4, and the outcomes of each configuration are addressed below:

5.1.1. Configuration-1: PV/wind/hydro/diesel/battery

The solar PV, wind, diesel generators, and batteries are the energy sources allocated in configuration 1 to meet the village energy demand. The system's different sizes—photovoltaic, diesel generator, wind turbine, hydro, and battery—are 93 kW, 100 Kw, 75 kW, 85 kW, and 50 numbers, respectively. The predicted annual energy demand is 463,925 kWh, and the availability of excess energy is 9.4 %. The net present cost (NPC) was \$1,910,382 and the cost of energy (COE) is 0.2102\$/KWh.

5.1.2. Configuration-2: PV/wind/diesel/battery

Configuration-2 takes into consideration solar energy, wind turbines, diesel engines, batteries excluding hydro. The system's energy sources sizing is 100 kW, 82 kW, 100 Kw, and 75 numbers for PV, wind, diesel, and batteries, respectively. The anticipated annual energy consumption is 462,987 kWh, and the potential supply of excess energy is 10.85 %. Additionally, the net present cost (NPC) is \$1,752,813 and the cost of energy (COE) is 0.1714\$/KWh.

5.1.3. Configuration-3: PV/wind/biogas/battery/diesel

Hydro is not taken into consideration in combination 3; instead, PV, wind, biogas, diesel, and battery systems are combined. PV, wind, biogas, diesel, and battery sizes are 100 kW, 95 KW, 100 KW, 100 KW,

Table 4

Configuration of Categorized system schemes using HOMER.

Architecture	PV Size (Kw)	Wind Turbine Size (Kw)	Diesel Genset Size (Kw)	Biogas Genset Size (Kw)	Hydro Genset Size (Kw)	Battery (No.)	COE (\$/kwh)	NPC (\$)
P-V./Wind/Hydro/Diesel/Battery	93	75	100	–	85	50	0.2102	1,910,382
PV/Wind/Diesel/Battery	100	82	100	–	–	75	0.1714	1,752,813
PV/Biogas/Battery/Diesel	100	95	100	100	–	76	0.1007	1,474,182
Biogas/Diesel	–	–	100	100	–	–	0.01842	1,198,997

and 76 units respectively. The anticipated energy demand for the system is 386,113 kWh/year, and the projected supply of excess energy is 16.75 %. In this setup, the net present cost (NPC) is \$1,474,182 and the cost of energy (COE) is \$0.1007\$/KWh.

5.1.4. Configuration-4: biogas/diesel

Diesel and biogas generators are considered in combination 4. The system's sizes are 100 kW for the biogas generator and 100 kW for the diesel generator. The anticipated annual energy demand is 34,177.2 kWh, and the projected amount of excess energy is 27.49 %. Additionally, the net present cost (NPC) was \$1,198,997 and the cost of energy (COE) was 0.01842/KWh.

The total cost breakdown of combination-4's constituent parts is displayed in Table 1. Biogas/Diesel configuration is shown to be the most economical solution having both the lowest cost of energy and NPC. It also produces the highest amount of excess energy which makes it suitable in cases of anticipated expansion. The battery backup system has the greatest overall cost of \$12,939 among several components, while the biogas generator gives the lowest cost at \$3975. This happens as a result of the abundance in availability of animal manure, thus conversely decreasing the need for diesel and photovoltaic systems, which are highly expensive. Thus, configurations 3 and 4 with biogas architecture have lower COE and NPC compared to configurations without it while configurations with battery have relatively higher cost.

5.2. Goa based simulation

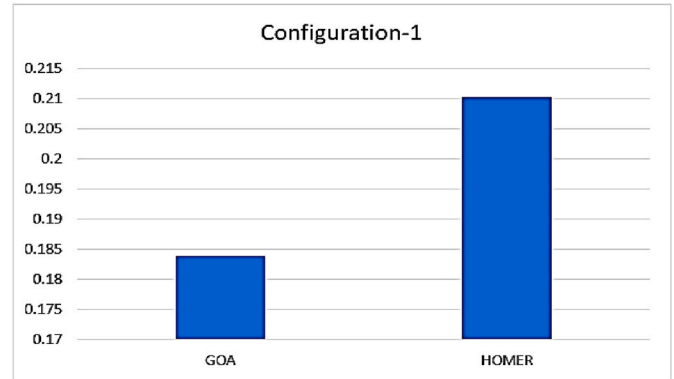
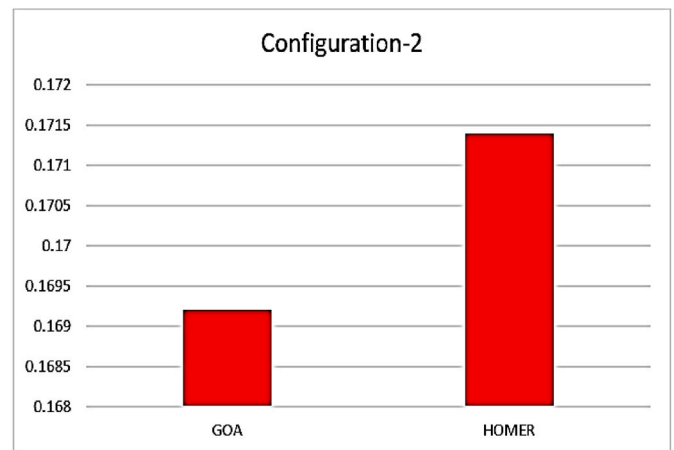
Just as in the HOMER-based simulation, Table 5 also lists four configurations of HRES variables that are taken into consideration. However, GOA was used for optimization, following the previously described process. GOA was used to optimize NPC, COE, and operational cost results in various combinations.

5.3. Comparison of results

The hourly simulation is used to calculate the hybrid power plant in both load-following and cycle-charging strategies. When the batteries are low, the diesel engine supplies the load. Alternatively, the solar PV supplies the load.

Tables 1 and 2 present the HOMER and GOA simulation findings.

Figs. 12–19 display the economic outcomes of various combinations of HRES using HOMER and GOA. GOA shows a better result for configurations 1 to 4 compared with HOMER. Of the four configurations considered under GOA, configuration-4 shows the lowest COE of 0.01783\$/kWh and minimal NPC of \$1,062,739. A 0 % capacity

**Fig. 12.** COE Configuration-1.**Fig. 13.** COE Configuration-2.

shortfall was also recorded as the system could fully meet the load demand of the village. The solar PV system generates 1,74,499 kWh of electricity annually. Due to the high cost of fuel in Nigeria, the cost of the diesel engine is higher than the cost of the battery at a total of 4,25,623 kWh/yr. The findings indicate that, in comparison to HOMER, GOA-based optimization is more economical. GOA's energy costs are 0.01783 \$/kWh, while HOMER's are 0.01842\$/kWh. Moreover, the GOA-based system has a higher renewable energy penetration rate than

Table 5

Configuration of Categorized system schemes using GOA.

Architecture	PV Size (Kw)	Wind Turbine Size (Kw)	Diesel Genset Size (Kw)	Biogas Genset Size (Kw)	Hydro Genset Size (Kw)	Battery (No.)	COE (\$/kwh)	NPC (\$)
P-V./Wind/Hydro/Diesel/Battery	86	73	88	–	62	36	0.1839	1,802,118
PV/Wind/Diesel/Battery	85	78	95	–	–	55	0.1692	1,519,423
PV/Biogas/Battery/Diesel	68	84	90	110	–	62	0.08461	1,362,109
Biogas/Diesel	–	–	120	95	–	–	0.01783	1,062,739

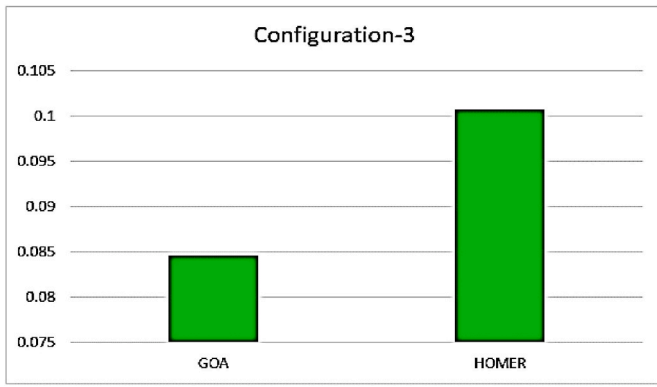


Fig. 14. COE Configuration-3.

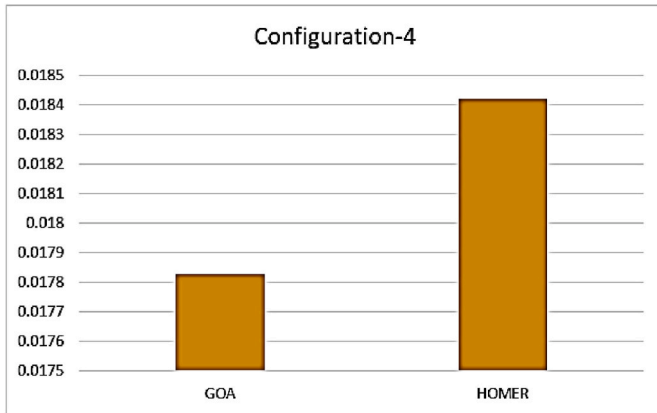


Fig. 15. COE Configuration-4.

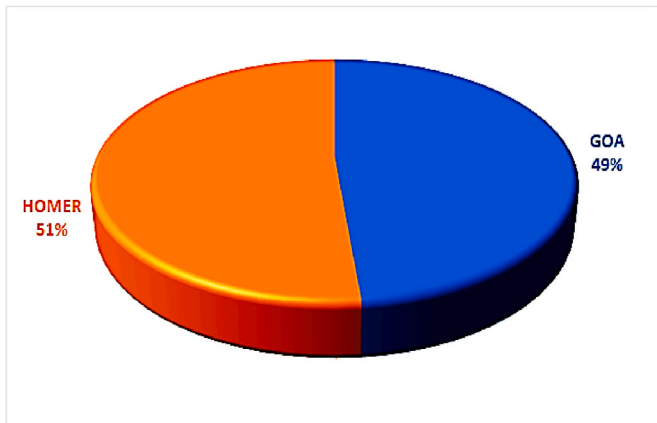


Fig. 16. NPC Configuration_1.

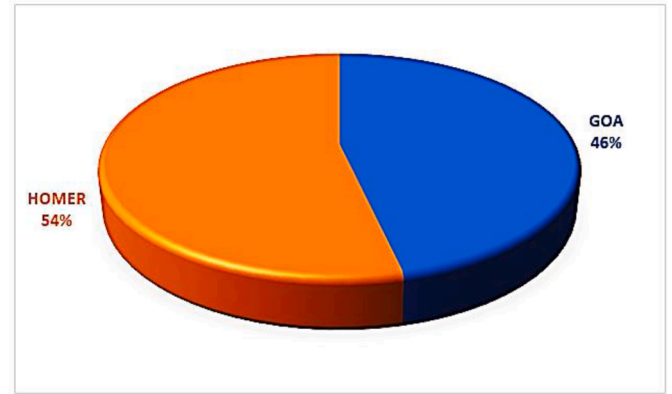


Fig. 17. NPC Configuration_2.

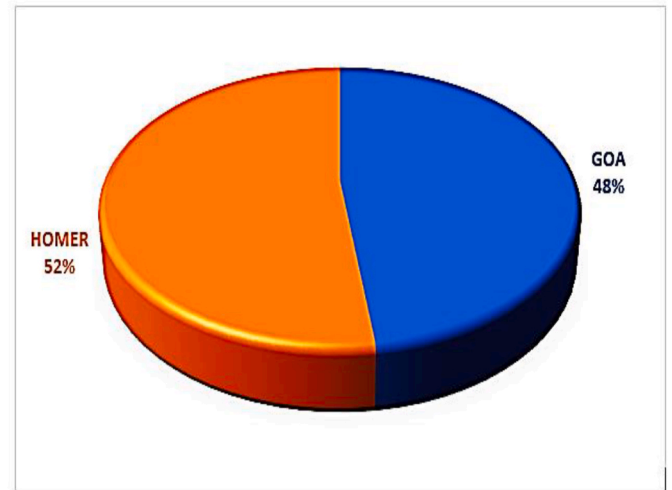


Fig. 18. NPC Configuration_3.

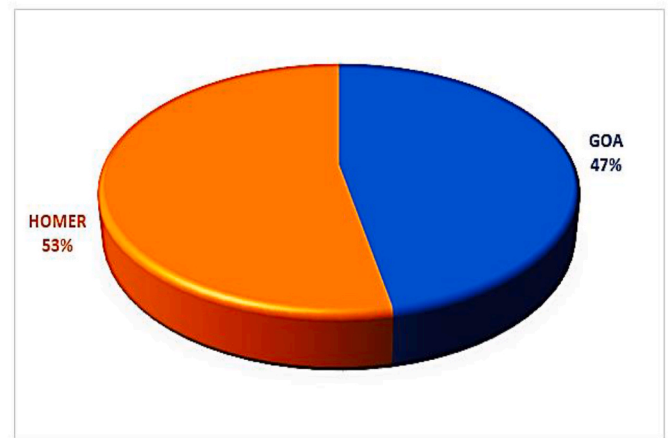


Fig. 19. NPC Configuration_4.

HOMER. In addition, GOA emits fewer CO₂ emissions than HOMER. The results show that the grasshopper optimization algorithm (GOA) produces the best results for COE and NPC. GOA also displays a relatively better capacity optimization of the microgrid components than HOMER as shown in Fig. 18 through 21 while still able to meet all load demand. In configuration 3, GOA preferred an increased capacity for biogas owing to the cheapness and abundant availability of biomass resources of the village, and reduced capacity for the other architectures due to the configuration higher cost. The biogas capacity proposed is just about 5 % higher than the peak load demand of the village. This ensures that excess electricity production is minimized while simultaneously meeting

demand. For the peak load demand of configuration 4, In contrast to HOMER, GOA predicted a more compact biogas architecture, which is just enough to meet the peak load demand thereby ensuring a minimal excess electricity production. The proposed capacity of biogas in Figs. 20 and 23 by GOA also indicates that in cases where there is abundant biomass, it is preferred to run the microgrid system solely on biogas and other energy resources as a backup. Furthermore, GOA has a quick

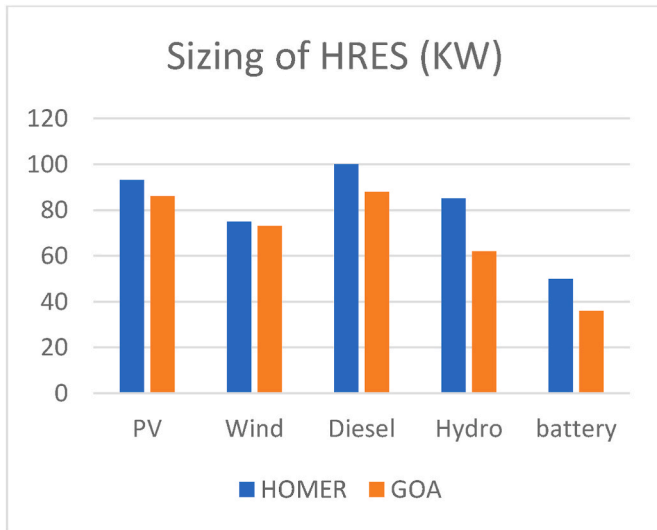


Fig. 20. Sizing of HRES Configuration_1.

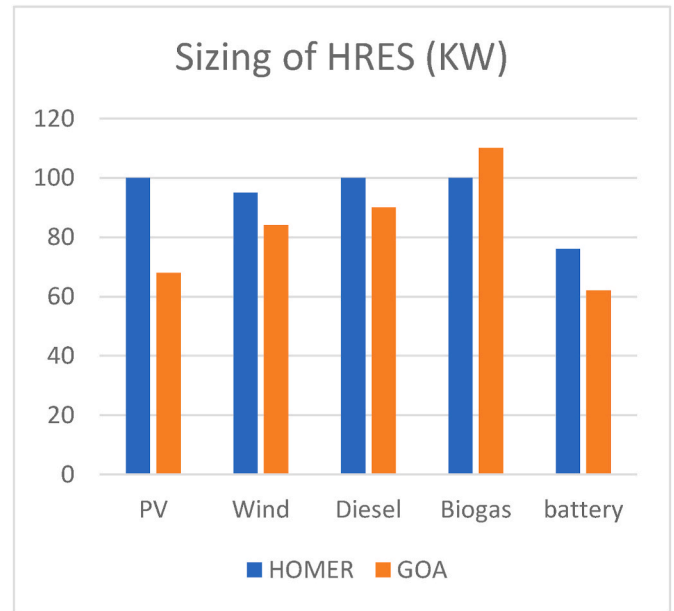


Fig. 22. Sizing of HRES Configuration_3.

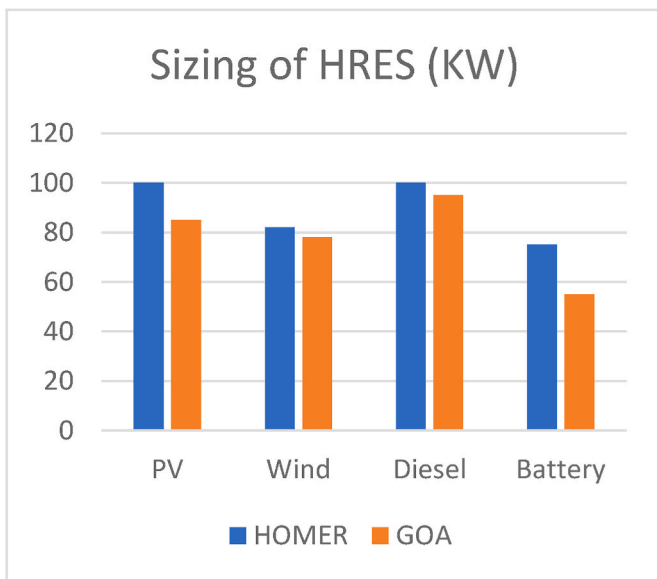


Fig. 21. Sizing of HRES Configuration_2.

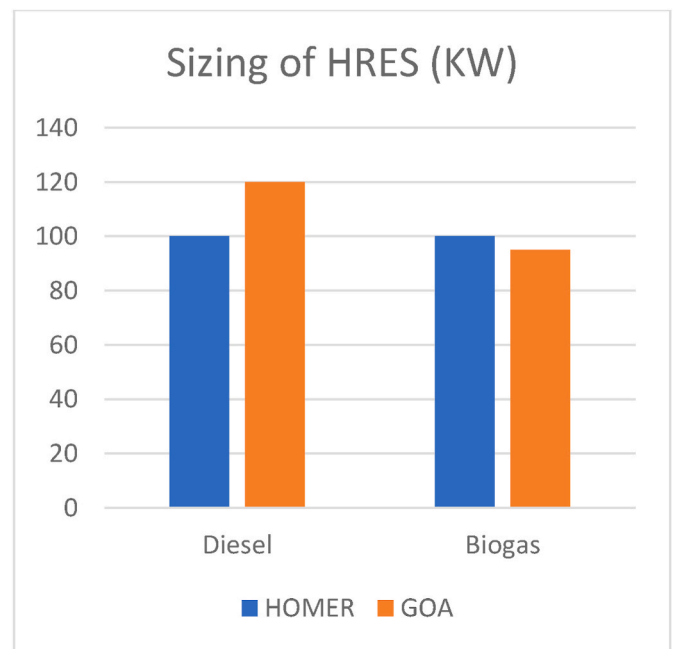


Fig. 23. Sizing of HRES Configuration_4.

convergence rate and can manage uncertainties in load demand, wind speed, and solar radiation. In addition, for both HOMER and GOA, configurations with biogas architecture have a better economic advantage. Also the higher the biogas penetration into the microgrid system, the lower the COE and NPC. This is because biogas is more affordable and nearly always available to meet load demand. This demonstrates that when biomass penetration rises, the impact of other energy sources on electricity generation will decrease. At a certain biomass level, the hybrid power plant will operate solely on biogas. Fig. 22 shows the Sizing of HRES Configuration 3.

Figs. 24–26 shows the Battery State of Charge (SOC) for configuration 1,2,3. To meet maximum demand or expected power outages during the night, the battery bank can drain with more power (approximately to the minimum state of charge). The battery bank can quickly charge and discharge during the year by anticipating future periods of extra electricity, which is crucial for lowering excess electricity levels. The initial price of PV systems has a more noticeable effect on the cost of energy (COE) because of the increasing solar potential and consequently larger PV installation capacity. The drop in the COE and NPC of renewable

energy for peak demands highlights the growing importance of biogas generators as an affordable local power supply to meet energy demands. The decision is made between using the diesel generator and draining the storage bank if the SOC exceeds the minimal value of 20 %. The less expensive option is given priority in providing energy to the remaining load, and if the first option is unable to meet demand, a second option begins to operate. The demand is met when overall production is equals to the primary load, and excess power is utilized to charge the battery bank or dumped.

6. Conclusion

The hybrid off-grid power system modeling and optimization for rural electrification using Nsukka as a case study were presented in this

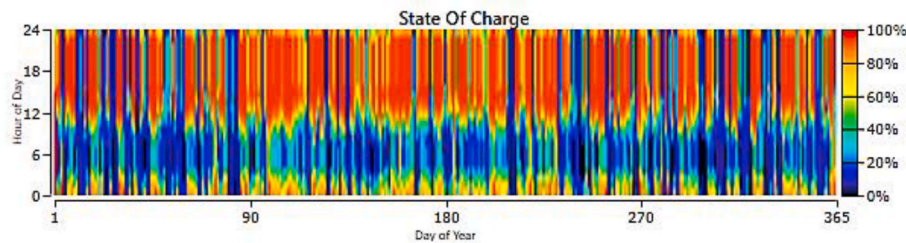


Fig. 24. Battery State of Charge (SOC) for configuration-1.

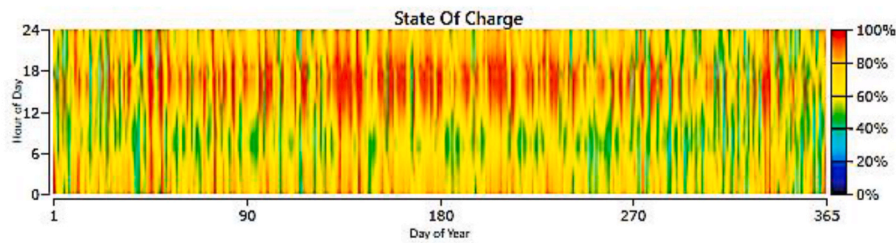


Fig. 25. Battery State of Charge (SOC) for configuration-2.

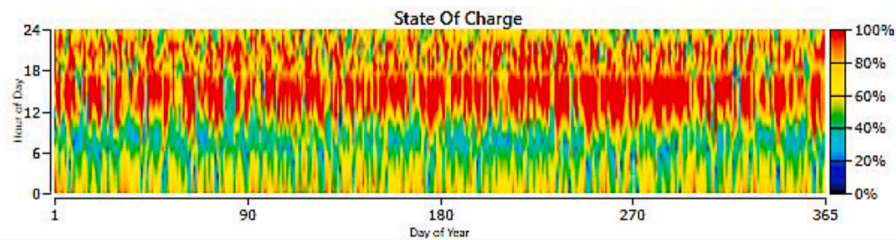


Fig. 26. Battery State of Charge (SOC) for configuration-3.

research. The system operating plan was developed by taking into account several criteria, including the system's emissions, source allocation, energy demand estimation, and economic aspect comparison. Additionally, using HOMER Pro and GOA software, four different hybrid energy system configurations have been assessed which include PV/Wind/Diesel/Battery, PV/Wind/Diesel/Battery, PV/Biogas/Diesel/Battery, and Biogas/Diesel. First, HOMER is assessed using the four possible HRES combinations. The Biogas/Diesel combination yields the lowest COE and minimum NPC of \$1,198,997 and 0.01842/kWh among these combinations. When the biomass fluctuates between 1 and 5 tones, Biogas/Diesel provides the lowest COE and minimal NPC of \$1,062,739 and 0.01783 \$/kWh. Additionally, a sensitivity evaluation is provided for variations in the annual wind speed, biomass amount, and flow rate. Lastly, the study includes a comparison analysis. When evaluating the four different HRES combinations utilizing HOMER and GOA, it is shown that GOA-based optimization is more economical than HOMER, with a lower COE of 0.01783 \$/kWh and without unmet load. In comparison to HOMER, the GOA-based system has lower CO₂ emissions and a higher penetration of Biogas and PV System. The biogas capacity proposed is just about 5 % higher than the peak load demand of the village. This ensures that excess electricity production is minimized while simultaneously meeting demand. For the peak load demand of configuration 4, In contrast to HOMER, GOA predicted a more compact biogas architecture, which is just enough to meet the peak load demand thereby ensuring a minimal excess electricity production.

7. Recommendation

When creating new meta-heuristic algorithms, future studies should

consider special qualities of MGs, such as complicated architecture, unpredictable generation and demand, and constraints. The efficiency of the algorithms can be increased by combining MHOAs with other methods. Research should also concentrate on enhancing the effectiveness and efficiency of MHOAs in MGs by promoting the development of novel search strategies and procedures for fine-tuning parameters, while also tackling the uncertainties present in MGs, such as the unpredictability of RES and demand load. The advancement of these research opportunities may greatly enhance the decision-making process in the operation and planning of MGs, resulting in a greater environment, society, and economic growth.

CRediT authorship contribution statement

Timothy Oluwaseun Araoye: Writing – original draft, Software, Project administration, Methodology, Formal analysis, Data curation, Conceptualization. **Evans Chinemezu Ashigwuike:** Validation, Supervision, Resources. **Muncho Josephine Mbunwe:** Writing – review & editing, Visualization, Validation, Investigation. **Oladipupo Idris Bakinson:** Writing – review & editing, Investigation, Formal analysis, Data curation. **ThankGod Izuchukwu Ozue:** Project administration, Investigation, Formal analysis, Data curation.

Declaration of competing interest

Authors declare no conflict of interest as regards to publication of this paper.

Acknowledgements

The authors acknowledge the support received from Energy and Sustainable Research Group (ESRG), University of Nigeria, Nsukka that

enabled the timely completion of this research. Also, the authors expressed their gratitude to the editor and reviewers of this journal whose feedback has been instrumental in enhancing this paper.

Appendix

Table A
Solar panel parameters [126].

Parameters	Value
Rated power	330 W
Maximum voltage at STC, V_m (STC)	41.9 V
Open circuit voltage, V_{oc}	50.9 V
Open circuit voltage at STC, V_{oc} (STC)	37.3 V
Maximum Current at STC, I_m (STC)	7.88A
Short-circuit current, I_{sc}	8.47A
Short-circuit current at STC, I_{sc} (STC)	8.78A
Current temperature coefficient, μ_{Isc}	$-173.1 \text{ mV}/^\circ\text{C}$
Voltage temperature coefficient, μ_{Voc}	$3.6 \text{ mA}/^\circ\text{C}$
PV empirical adjustment parameter, σ_{oc}	-0.04
Thermal Voltage, V_{th}	25.7 Mv
Cell Temperature at STC, T_c (STC)	25 $^\circ\text{C}$
Irradiance at STC, G (STC)	1000 W/m^2

Table B
Wind turbine parameters.

Parameters	Value
Rated power	1 kW
Rated voltage	110 V
Startup wind speed	1.5 m/s
Maximum wind speed	10 m/s
Rated wind speed	50 m/s
Rated rotation speed	300U/min
Fan blade quantity	3
Rotor blades diameter	2 m
Tip speed	31 m/s

Table C
Battery parameters.

Parameters	Value
Type	Lithium-ion
Fully charge voltage	14 V
Capacity	3000 Ah
Charging current	2A–10A
Charging Voltage-	14.4–14.9 V
Float voltage	13.5–13.8 V
I_{rate}	6C

Table D
Hydropower Parameters

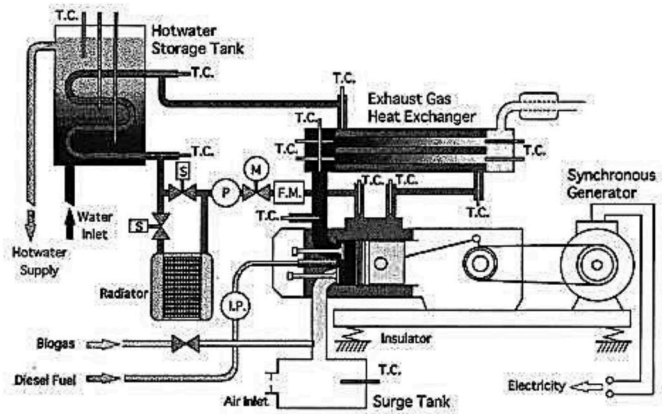
Parameters	Value
Rated power	20kw
Rated Head	23m
Rated Flow	142L/s
Penstock diameter	300 mm
Rated voltage	440V
Rated Current	35A
Rated Speed	1200r/min
Ambient Temperature	$-25^\circ\text{C} - +50^\circ\text{C}$
Efficiency	80 %

Table E
Generator Engine Specification

Generator Specification	Description
Genset Manufacturer	Greaves power
Genset Rating	Prime power
Genset output (KVA/KW)	125/100
Genset model	Gpwill-pll-125
Engine rating or max. Power at rated RPM in kw (BHP)	114 (155)
Engine Model	4G11TAG26
Engine type	TCAC
No. of cylinders/cylinder arrangement	4/inline
Bore and stroke (mm)	108 x 133
Compression ratio	16:8
Rated RPM	1500
Governor: Type/class of Governing	Mechanical/G2
Over speed (rpm)	1650
Air cleaner type/Qty	Dry/01
Fuel tank capacity (L)	350



Appendix 1Generator operating on 100 % Biogas system with electronic governing ignition system [127]



Appendix 2. A hybrid micro-grid system based on diesel fuel and Biogas [127].

References

[1] Suresh Vendoti, M. Muralidhar, R.R. Kiranmayi, Performance analysis of hybrid power system along with conventional energy sources for sustainable development in rural areas, *Int. J. Recent Technol. Eng.* 8 (3) (2019) 5971–5977.

[2] S. Rajanna, R.P. Saini, Development of optimal integrated renewable energy model with battery storage for a remote Indian area, *Energy* 111 (2016) 803–817.

[3] A. Chauhan, R.P. Saini, Discrete harmony search based size optimization of integrated renewable energy system for remote rural areas of Uttarakhand and state in India, *Renew. Energy* 94 (2016) 587–604.

[4] A. Chauhan, R.P. Saini, Techno-economic optimization based approach for energy management of a stand-alone integrated renewable energy system for remote areas of India, *Energy* 94 (2016) 138–156.

[5] Lanre Olatomiwa, Richard Blanchard, Saa Mekhilef, Daniel Akinyele, Hybrid renewable energy supply for rural healthcare facilities: an approach to quality healthcare delivery, *Sustain. Energy Technol. Assess.* 30 (2018) 121–138.

[6] Lanre Olatomiwa, Saad Mekhilef, A.S.N. Huda, Olayinka S. Ohunakin, Economic evaluation of hybrid energy systems for rural electrification in six geo-political zones of Nigeria, *Renew. Energy* 83 (2015) 435–446.

[7] Lanre Olatomiwa, Saad Mekhilef, A.S.N. Huda, Kamilu Sanusi, Techno-economic analysis of hybrid PV–diesel–battery and PV–wind–diesel–battery power systems for mobile BTS: the way forward for rural development, *Energy Sci. Eng.* 3 (4) (2015) 271–285.

[8] Mohamed A. Mohamed, Ali M. Eltamaly, Abdulrahman I. Alolah, PSO-based smart grid application for sizing and optimization of hybrid renewable energy systems, *PLoS One* 11 (8) (2016).

[9] Monotosh Das, Kumar Singh, Maisanam Anil, Agnimitra Biswas, Techno-economic optimization of an off-grid hybrid renewable energy system using meta-heuristic optimization approaches of a radio transmitter station in India, *Energy Convers. Manag.* 185 (2019) 339–352.

[10] Weiping Zhang, Akbar Maleki, Marc A. Rosen, Jingqing Liu, Sizing a stand-alone solar-wind-hydrogen energy system using weather forecasting and a hybrid search optimization algorithm, *Energy Convers. Manag.* 180 (2019) 609–621.

[11] M.M. Samy, S. Barakat, H.S. Ramadan, A flower pollination optimization algorithm for an off-grid PV-fuel cell hybrid renewable system, *Int. J. Hydrogen Energy* 44 (2018) 2141–2152.

[12] S. Mahapatra, S. Dasappa, Energy for sustainable development rural electrification: optimising the choice between decentralized renewable energy sources and grid extension, *Energy Sustain Dev* 16 (2) (2012) 146–154.

- [13] M. Bajaj, A. Kumar Singh, Design and analysis of optimal passive filters for increasing the harmonic-constrained hosting capacity of inverter-based DG systems in non-sinusoidal grids, *Electr. Eng.* 104 (2021) 1883–1907.
- [14] S. Munuswamy, K. Nakamura, A. Katta, Comparing the cost of electricity sourced from a fuel cell-based renewable energy system and the national grid to electrify a rural health Centre in India: a case study, *Renew. Energy* 36 (11) (2011) 2978–2983.
- [15] M. Fadaeenejad, M.A.M. Radzi, M.Z.A. AbKadir, H. Hizam, Assessment of hybrid renewable power sources for rural electrification in Malaysia, *Renew. Sustain. Energy Rev.* 30 (2014) 299–305.
- [16] R. Rajbongshi, D. Borgohain, S. Mahapatra, Optimization of PV-biomass-diesel and grid base hybrid energy systems for rural electrification by using HOMER, *Energy* 126 (2017) 461–474.
- [17] S. Sharma, Y.R. Sood, Optimal planning, and sensitivity analysis of green microgrid using various types of storage systems, *Wind Eng.* 45 (4) (2021) 939–952.
- [18] S. Choudhury, S.K. Acharya, R.K. Khadanga, S. Mohanty, J. Arshad, A. Ur Rehman, et al., Harmonic profile enhancement of grid connected fuel cell through cascaded H-bridge multi-level inverter and improved squirrel search optimization technique, *Energies* 14 (2021) 7947.
- [19] A. Yousef, R.M. Asif, M. Shakir, A.U. Rehman, F. Alassery, H. Hamam, et al., A novel machine learning-based price forecasting for energy management systems, *Sustainability* 13 (22) (2021) 12693.
- [20] A. Yousef, R.M. Asif, M. Shakir, A.U. Rehman, M.S. Adrees, An improved residential electricity load forecasting using a machine-learning-based feature selection approach and a proposed integration strategy, *Sustainability* 13 (2021) 6199.
- [21] A. Khan, M.H. Jaffery, Y. Javed, J. Arshad, A.U. Rehman, R. Khan, et al., Hardware-in-the-Loop implementation and performance evaluation of three-phase hybrid shunt active power filter for power quality improvement, *Math. Probl. Eng.* 1–23 (2021).
- [22] M.A. Bakar Siddique, A. Adeel, R.M. Asif, A.U. Rehman, M.T. Sadiq, I. Ullah, Implementation of incremental conductance MPPT algorithm with integral regulator by using boost converter in grid-connected PV array, *IETE J. Res.* 69 (2021) 3822–3835.
- [23] Inci Mustafa, Aygen Mehmet Selim, A modified energy management scheme to support phase balancing in grid-interfaced photovoltaic/fuel cell systems, *Ain Shams Eng. J.* 12 (3) (2021) 2809–2822. ISSN 2090-4479.
- [24] S. Kumar, T. Kaur, S. Upadhyay, V. Sharma, D. Vatsal, Optimal sizing of stand-alone hybrid renewable energy system with load shifting, *Energy Sources Part A* (2020) 1–20.
- [25] J.O. Oladigbolu, M.A.M. Raml, Y.A. Al-Turki, Optimal design of a hybrid PV solar/micro-hydro/diesel/battery energy system for a remote rural village under tropical climate conditions, *Electronics* 9 (9) (2020) 1491.
- [26] Bhattacharjee Somudeep, Nandi Champa, Design of a voting-based smart energy management system of the renewable energy-based hybrid energy system for a small community, *Energy* 214 (2021) 118977. ISSN 0360-5442.
- [27] P. Arévalo, F. Jurado, Performance analysis of a PV/HKT/WT/DG hybrid autonomous grid, *Electr. Eng.* 103 (1) (2021) 227–244.
- [28] H. El-Khoury, A. Allouhi, T. Salameh, T. Kousksou, A. Jamil, B. El Amrani, Energy, Economic, Environment (3E) analysis of WT-PV-Battery autonomous hybrid power plants in climatically varying regions, *Sustain. Energy. Technol. Assess.* 43 (2021) 100961. ISSN 2213-1388.
- [29] S. Upadhyay, M.P. Sharma, Development of hybrid energy system with cycle charging strategy using particle swarm optimization for a remote area in India, *Renew. Energy* 77 (2015) 586–598.
- [30] S. Upadhyay, M.P. Sharma, Selection of a suitable energy management strategy for a hybrid energy system in a remote rural area of India, *Energy* 94 (2016) 352–366.
- [31] S. Singh, M. Singh, S.C. Kaushik, Optimal sizing of grid integrated hybrid PV-biomass energy system using artificial bee colony algorithm, *IET Renew. Power Gener.* 10 (5) (2016) 642–650.
- [32] A. González, J.R. Riba, A. Rius, et al., Optimal sizing of a hybrid grid-connected photovoltaic and wind power system, *Appl. Energy* 154 (2015) 752–762.
- [33] M. Zabarjadi, A. Askarzadeh, Optimization of a reliable grid-connected PV-based power plant with/without energy storage system by a heuristic approach, *Sol. Energy* 25 (2016) 12–21.
- [34] M.A. Mohamed, A.M. Eltamaly, A.I. Alolah, Swarm intelligence-based optimization of grid-dependent hybrid renewable energy systems, *Renew. Sustain. Energy Rev.* 77 (2017) 515–524.
- [35] Y. Sawle, S.C. Gupta, A.K. Bohre, Socio-techno-economic design of hybrid renewable energy system using optimization techniques, *Renew. Energy* 119 (2018) 459–472.
- [36] A.M. Abdelshafya, H. Hassana, J. Jakub, Optimal design of a grid-connected desalination plant powered by renewable energy resources using a hybrid PSO-GWO approach, *Energy Convers. Manag.* 173 (2018) 331–347.
- [37] A. Gupta, R.P. Saini, M.P. Sharma, Steady-state modeling of the hybrid energy system for off-grid electrification of a cluster of villages, *Renew. Energy* 35 (1) (2010) 520e35.
- [38] Uddin Moslem, Huadong Mo, Daoyi Dong Sondoss Elsawah, Techno-economic potential of multi-energy community microgrid: the perspective of Australia, *Renew. Energy* 219 (2023) 119544.
- [39] C.A. Sima, C.L. Popescu, M.O. Popescu, M. Roscia, G. Seritan, C. Panait, Technoeconomic assessment of university energy communities with on/off microgrid, *Renew. Energy* 193 (2022) 538–553.
- [40] F. Vivas, A. De Las Heras, F. Segura, J.M. Andújar, A review of energy management strategies for renewable hybrid energy systems with hydrogen backup, *Renew. Sustain. Energy Rev.* 82 (2018) 126–155.
- [41] A. Khan, T.A. Alghamdi, Z.A. Khan, A. Fatima, S. Abid, A. Khalid, N. Javaid, Enhanced evolutionary sizing algorithms for optimal sizing of a Stand-Alone PV-WT-Battery hybrid system, *Appl. Sci.* 9 (2019) 5197.
- [42] C. Ammari, D. Belatrache, B. Touhami, S. Makhloufi, Sizing, optimization, control and energy management of hybrid renewable energy system—a review, *Energy Built Environ* 3 (2002) 399–411.
- [43] B. Modu, P. Abdullah, A.L. Bukar, M.F. Hamza, A systematic review of hybrid renewable energy systems with hydrogen storage: sizing, optimization, and energy management strategy, *Int. J. Hydrogen Energy* 48 (2023) 38354–38373.
- [44] A.A. Khan, A.F. Minai, R.K. Pachauri, H. Malik, Optimal sizing, control, and management strategies for hybrid renewable energy systems: a comprehensive review, *Energies* 15 (2022) 6249.
- [45] T. Afolabi, H. Farzaneh, Optimal design and operation of an off-grid hybrid renewable energy system in Nigeria's rural residential area, using fuzzy logic and optimization techniques, *Sustainability* 15 (2023) 3862.
- [46] B. Barakat, M.M. Samy, M.B. Eteiba, W.I. Wahba, Viability study of grid connected PV/Wind/Biomass hybrid energy system for a small village in Egypt, in: 2016 Eighteenth International Middle East Power Systems Conference (MEPCON), Cairo, Egypt, 2016, pp. 46–51.
- [47] N. Alshammari, M.M. Samy, J. Asumadu, Optimal economic analysis study for renewable energy systems to electrify remote region in kingdom of Saudi Arabia, in: 2018 Twentieth International Middle East Power Systems Conference (MEPCON), Cairo, Egypt, 2018, pp. 1040–1045, <https://doi.org/10.1109/MEPCON.2018.8635287>.
- [48] Wind Europe, Renewable Hybrid Power Plants-Exploring the Benefits and Market Opportunities, 2019. Available online: <https://windeurope.org/wp-content/uploads/files/policy/position-papers/WindEurope-renewable-hybrid-power-plants-as-benefits-and-market-opportunities.pdf>. (Accessed 30 April 2024).
- [49] O. Lindberg, J. Arqvist, J. Munkhammar, D. Lingfors, Review on power-production modeling of hybrid wind and PV power parks, *J. Renew. Sustain. Energy* 13 (2021) 042702.
- [50] C. Chang, Multi-choice goal programming model for the optimal location of renewable energy facilities, *Renew. Sustain. Energy Rev.* 41 (2015) 379–389.
- [51] R. Wang, G. Li, M. Ming, G. Wu, L. Wang, An efficient multi-objective model and algorithm for sizing a stand-alone hybrid renewable energy system, *Energy* 141 (2017) 2288–2299.
- [52] A. Khan, M. Naeem, M. Iqbal, S. Qaisar, A. Anpalagan, A compendium of optimization objectives, constraints, tools and algorithms for energy management in microgrids, *Renew. Sustain. Energy Rev.* 58 (2016) 1664–1683.
- [53] L. Moretti, M. Astolfi, C. Vergara, E. Macchi, J.I. Pérez-Arriaga, G. Manzolini, A design and dispatch optimization algorithm based on mixed integer linear programming for rural electrification, *Appl. Energy* 233–234 (2019) 1104–1121.
- [54] N. Wu, H. Wang, Deep learning adaptive dynamic programming for real time energy management and control strategy of micro-grid, *J. Clean. Prod.* 204 (2018) 1169–1177.
- [55] R. Siddaiah, R. Saini, A review on planning, configurations, modeling and optimization techniques of hybrid renewable energy systems for off grid applications, *Renew. Sustain. Energy Rev.* 58 (2016) 376–396.
- [56] B. Das, A. Kumar, A NLP approach to optimally size an energy storage system for proper utilization of renewable energy sources, *Procedia Comput. Sci.* 125 (2018) 483–491.
- [57] S.S. Reddy, J.A. Momoh, Realistic and transparent optimum scheduling strategy for hybrid power system, *IEEE Trans. Smart Grid* 6 (2015) 3114–3125.
- [58] A.P.J. Stanley, J. King, Optimizing the physical design and layout of a resilient wind, solar, and storage hybrid power plant, *Appl. Energy* 317 (2022) 119139.
- [59] D. Abbes, A. Martinez, G. Champenois, Life cycle cost, embodied energy and loss of power supply probability for the optimal design of hybrid power systems, *Math. Comput. Simulat.* 98 (2014) 46–62.
- [60] P. Roy, J. He, Y. Liao, Cost minimization of Battery-Supercapacitor hybrid energy storage for hourly dispatching Wind-Solar hybrid power system, *IEEE Access* 8 (2020) 210099–210115.
- [61] Y. Sawle, S. Gupta, A.K. Bohre, Techno-economic scrutiny of HRES through GA and PSO technique, *Int. J. Renew. Energy Technol.* 9 (84) (2018).
- [62] N. Muleta, A.Q.H. Badar, Designing of an optimal standalone hybrid renewable energy micro-grid model through different algorithms, *J. Eng. Res.* 11 (2023) 100011.
- [63] M. Sharafi, T.Y. ElMekkawy, Multi-objective optimal design of hybrid renewable energy systems using PSO-simulation based approach, *Renew. Energy* 68 (2014) 67–79.
- [64] L. Petersen, F. Iov, G.C. Tarnowski, V. Gevorgian, P. Koralewicz, D. Stroe, Validating performance models for hybrid power plant control assessment, *Energies* 12 (2019) 4330.
- [65] K. Anoune, M. Bouya, A. Astito, A.B. Abdellah, Sizing methods and optimization techniques for PV-wind based hybrid renewable energy system: a review, *Renew. Sustain. Energy Rev.* 93 (2018) 652–673.
- [66] S. Upadhyay, M. Sharma, A review on configurations, control and sizing methodologies of hybrid energy systems, *Renew. Sustain. Energy Rev.* 38 (2014) 47–63.
- [67] S.O. Bade, A. Meenakshisundaram, O.S. Tomomewo, Current status, sizing methodologies, optimization techniques, and energy management and control strategies for Co-located utility-scale wind-solar-based hybrid power plants: a review, *Eng* 5 (2024) 677–719.

- [68] A. Chauhan, R. Saini, A review on Integrated Renewable Energy System based power generation for stand-alone applications: configurations, storage options, sizing methodologies and control, *Renew. Sustain. Energy Rev.* 38 (2014) 99–120.
- [69] P. Ganguly, A. Kalam, A. Zayegh, Solar–wind hybrid renewable energy system: current status of research on configurations, control, and sizing methodologies, in: *Hybrid-Renewable Energy Systems in Microgrids*, Elsevier, Amsterdam, The Netherlands, 2018, pp. 219–248.
- [70] J. Lian, Y. Zhang, C. Ma, Y. Yang, E. Chaima, A review on recent sizing methodologies of hybrid renewable energy systems, *Energy Convers. Manag.* 199 (2019) 112027.
- [71] C.D. Iweh, S.G. Clarence, A.H. Roger, The optimization of hybrid renewables for rural electrification: techniques and the design problem, *Int. J. Eng. Trends Technol.* 70 (2022) 222–239.
- [72] M. Riaz, S. Ahmad, I. Hussain, M. Naeem, L. Mihet-Popa, Probabilistic optimization techniques in smart power system, *Energies* 15 (2022) 825.
- [73] M. Dubey, V. Kumar, M. Kaur, T.P. Dao, A systematic review on harmony search algorithm: theory, literature, and applications, *Math. Probl Eng.* (2021) 5594267, 2021.
- [74] A.G. Gad, Particle swarm optimization algorithm and its applications: a systematic review, *Arch. Comput. Methods Eng.* 29 (2022) 2531–2561.
- [75] Y. Wang, T. Wang, S. Dong, C. Yao, An improved grey-wolf optimization algorithm based on circle map, *J. Physics Conf. Ser.* 1682 (2020) 012020.
- [76] A. Gupta, S. Srivastava, Comparative analysis of ant colony and particle swarm optimization algorithms for distance optimization, *Procedia Comput. Sci.* 173 (2020) 245–253.
- [77] D. Shen, W. Ming, X. Ren, Z. Xie, Y. Zhang, X. Liu, A cuckoo search algorithm using improved beta distributing and its application in the process of edm, *Crystals* 11 (2021) 916.
- [78] N. Saiprasad, A. Kalam, A. Zayegh, Comparative study of optimization of HRES using HOMER and iHOGA software, *J. Sci. Ind. Res.* 77 (2018) 677–683.
- [79] M.S. Ramli, S.S.A. Wahid, K.K. Hassan, A comparison of renewable energy technologies using two simulation softwares: HOMER and RETScreen, *AIP Conf. Proc.* 1875 (2017) 030013.
- [80] A. Sari, A. Majidi, M.J.C. Oplencia, A. Timoshin, D.T.N. Huy, N.D. Trung, F. Alsaikhan, A.T. Hammid, A. Akhmedov, New optimized configuration for a hybrid PV/diesel/battery system based on coyote optimization algorithm: a case study for Hotan county, *Energy Rep.* 8 (2022) 15480–15492.
- [81] Y.-P. Xu, P. Ouyang, S.-M. Xing, L.-Y. Qi, M. Khayatnezhad, H. Jafari, Optimal structure design of a PV/FC HRES using amended Water Strider Algorithm, *Energy Rep.* 7 (2021) 2057–2067.
- [82] N. Ganjei, F. Zishan, R. Alayi, H. Samadi, M. Jahangiri, R. Kumar, A. Mohammadian, Designing and sensitivity analysis of an off-grid Hybrid Wind-solar power plant with diesel generator and battery backup for the rural area in Iran, *J. Eng.* (2022) 4966761, 2022.
- [83] K. Almutairi, S.S. Hosseini Dehshiri, S.J. Hosseini Dehshiri, A. Mostafaeipour, A. Issakhov, K. Techato, Use of a hybrid wind–solar–diesel–battery energy system to power buildings in remote areas: a case study, *Sustainability* 13 (2021) 8764.
- [84] A. Khosravani, E. Safaei, M. Reynolds, K.E. Kelly, K.M. Powell, Challenges of reaching high renewable fractions in hybrid renewable energy systems, *Energy Rep.* 9 (2023) 1000–1017.
- [85] B.J. Abdelhak, N. Essounbouli, A. Hamzaoui, F. Hnaïen, F. Yalaoui, Optimum sizing of hybrid PV/wind/battery using Fuzzy-Adaptive Genetic Algorithm in real and average battery service life, in: *Proceedings of the 2014 International Symposium on Power Electronics, Electrical Drives, Automation and Motion*, June 2014, pp. 18–20, Ischia, Italy.
- [86] A. Fetanat, E. Khorasanejad, Size optimization for hybrid photovoltaic-wind energy system using ant colony optimization for continuous domains-based integer programming, *Appl. Soft Comput.* 31 (2015) 196–209.
- [87] S. Ahmadi, S. Abdi, Application of the hybrid big bang-big crunch algorithm for optimal sizing of a stand-alone hybrid pv/wind/battery system, *Sol. Energy* 134 (2016) 366–374.
- [88] S. Sanajaoba, E. Fernandez, Maiden application of Cuckoo Search algorithm for optimal sizing of a remote hybrid renewable energy System, *Renew. Energy* 96 (2016) 1–10.
- [89] W. Peng, A. Maleki, M.A. Rosen, P. Azarikah, Optimization of a hybrid system for solar-wind-based water desalination by reverse osmosis comparison of approaches, *Desalination* 442 (2018) 16–31.
- [90] A. Khan, T.A. Alghamdi, Z.A. Khan, A. Fatima, S. Abid, A. Khalid, N. Javaid, Enhanced evolutionary sizing algorithms for optimal sizing of a Stand-Alone PV-WT-Battery hybrid system, *Appl. Sci.* 9 (2019) 5197.
- [91] A. Khan, N. Javaid, Jaya learning-based optimization for optimal sizing of stand-alone photovoltaic, wind turbine, and battery systems, *Engineering* 6 (2020) 812–826.
- [92] M. Javed, J. Jurasz, S. Ahmed, J. Mikulik, Performance comparison of heuristic algorithms for optimization of hybrid off-grid renewable energy systems, *Energy* 210 (2020) 118599.
- [93] M.T. Guneser, A. Elbaz, C. Seker, Hybrid optimization methods application on sizing and solving the economic dispatch problems of hybrid renewable power systems, in: *Applications of Nature-Inspired Computing in Renewable Energy Systems*, IGI Global, Hershey, PA, USA, 2022, pp. 136–165.
- [94] M. Jahannosh, S.A. Nowdeh, A. Naderipour, H. Kamyab, I. Faraji Davoudkhani, J. Jaromir Klemes, New hybrid meta-heuristic algorithm for reliable and cost-effective designing of photovoltaic/wind/fuel cell energy system considering load interruption probability, *J. Clean. Prod.* 278 (2021) 123406.
- [95] G. Ma, X. Guchao, Y. Chen, J. Rong, Multi-objective optimal configuration method for a standalone wind-solar-battery hybrid power system, *IET Renew. Power Gener.* 11 (2016) 194–202.
- [96] K.J. Senthil, R.S. Charles, D. Srinivasan, P. Venkatesh, Hybrid renewable energy-based distribution system for seasonal load variations, *Int. J. Energy Res.* 42 (2018) 1066–1087.
- [97] J. Radosavljevic, N. Arsic, M. Milovanovic, A. Ktena, Optimal placement and sizing of renewable distributed generation using hybrid metaheuristic algorithm, *J. Mod. Power Syst. Clean Energ.* 8 (2020) 499–510.
- [98] M.J. Aliabadi, M. Radmehr, Optimization of hybrid renewable energy system in radial distribution networks considering uncertainty using meta-heuristic crow search algorithm, *Appl. Soft Comput.* 107 (2021) 107384.
- [99] A.S. Alkafaji, A.A. Al-Samawi, H. Trabelsi, Hybrid energy storage review for renewable energy system technologies and applications, in: *Proceedings of the 2021 18th International Multi-Conference on Systems, Signals & Devices (SSD)*, 2021, pp. 1059–1067. Monastir, Tunisia, 22–25.
- [100] M. Stecca, L.R. Elizondo, T.B. Soeiro, P. Bauer, P. Palensky, A comprehensive review of the integration of battery energy storage systems into distribution networks, *IEEE Open J. Ind. Electron. Soc.* 1 (2020) 46–65.
- [101] A.M. Rabi, J. Radulovic, J.M. Buick, Comprehensive review of compressed air energy storage (CAES) technologies, *Thermoelectrics* 3 (2023) 104–126.
- [102] H. Xu, T. Yu, C. Chen, X. Wu, State-of-Charge prediction model for Ni-Cd batteries considering temperature and noise, *Appl. Sci.* 13 (2023) 6494.
- [103] F. Ishraque, S.A. Shezan, J. Nur, S. Islam, Optimal sizing and performance investigation of a solar-wind-battery-DG based hybrid Microgrid system applicable to the remote school of Bangladesh, *Eng. Rep.* 3 (2020) e12281.
- [104] K.M. Venkatachalam, V. Saravanan, Techno economic environmental assessment of hybrid renewable energy system in India, *Int. J. Adv. Appl. Sci.* 10 (2021) 343–362.
- [105] A. Khosravani, E. Safaei, M. Reynolds, K.E. Kelly, K.M. Powell, Challenges of reaching high renewable fractions in hybrid renewable energy systems, *Energy Rep.* 9 (2023) 1000–1017.
- [106] Muhammad Bakr Abdelghany, Ahmed Al-Durra, Hatem Zeineldin, Jiefeng Hu, Integration of cascaded coordinated rolling horizon control for output power smoothing in islanded wind–solar microgrid with multiple hydrogen storage tanks, *Energy* 291 (2024) 130442.
- [107] J. Xu, J. Wang, F. Wang, J. Wang, G. Li, SOC balancing control based on predictive power model amongst supercapacitor packs in MMC with embedded energy storage system, *IEEE Trans. Power Deliv.* 38 (4) (Aug. 2023) 2641–2649.
- [108] M. Combe, A. Mahmoudi, M.H. Haque, R. Khezri, Cost-effective sizing of an AC mini-grid hybrid power system for a remote area in South Australia, *IET Gener. Transm. Distrib.* 13 (2018) 277–287.
- [109] M. Thirunavukkarasu, Y. Sawle, A comparative study of the optimal sizing and management of off-grid solar/wind/diesel and battery energy systems for remote areas, *Front. Energy Res.* 9 (2021) 752043.
- [110] J.C. León Gómez, S.E. De León Aldaco, J. Aguayo Alquicira, A review of hybrid renewable energy systems: architectures, battery systems, and optimization techniques, *Eng* 4 (2023) 1446–1467.
- [111] Seman Soheyl, Mohamad Hossein Shafinei Mayam, Mehri Mehrjoo, Modeling a novel CCHP system including solar and wind renewable energy resources and sizing by a CC-MOPSO algorithm, *Appl. Energy* 184 (issues C) (2016) pg375–395.
- [112] Anurag Chauhan, R.P. Saini, Size optimization and demand response of a stand-alone integrated renewable energy system, *Energy* 124 (2017) 59–73.
- [113] A.B. Kanase-Patil, R.P. Saini, M.P. Sharma, Sizing of integrated renewable energy system based on load profile and reliability index for the state of uttarakh and in India, *Renew. Energy* 36 (11) (2011) 2809–2821.
- [114] R.E. Houssein, A. Yusuf, S. Mohammed, M. Saad, A. Makbul, Sizing of hybrid PV/Battery/Wind/Diesel microgrid system using an improved evolutionary algorithm considering uncertainties and battery degradation, *Sustainability* 15 (14) (2023).
- [115] Emile Niringiyimana, WanQuan Sun, Giovanni Dushimimana, Feasibility study of a hybrid PV/hydro system for remote area electrification in Rwanda, *J. Renewable Energy* (2022) 11, <https://doi.org/10.1155/2022/4030369>. Article ID 4030369.
- [116] M.Y. El-Sharkh, M. Tanrioven, A. Rahman, M.S. Alam, Cost related sensitivity analysis for optimal operation of a grid-parallel PEM fuel cell power plant, *J. Power Sources* 161 (2006) 1198–1207.
- [117] Suresh Vendoti, M. Muralidhar, R. Kiranmayi, Optimization of hybrid renewable energy systems for sustainable and economical power supply at SVCET Chittoor, i-manager's. *J. Power Syst. Eng.* 5 (11) (2017) 26–34.
- [118] S. Rajanna, R.P. Saini, Development of optimal integrated renewable energy model with battery storage for a remote Indian area, *Energy* 111 (2016) 803–817.
- [119] Nader Barsoum, Pearl Dianne Petrus, Cost optimization of hybrid solar, micro-hydro, and hydrogen fuel cells using Homer software, *Energy Power Eng.* 7 (2015) 337–347.
- [120] Arabi Nowdeh Saber, Hajibeigy Mahdi, Economic designing of PV/FC/Wind hybrid system considering components availability, *Int. J. Mod. Educ. Comput. Sci.* 7 (2013) 69–77.
- [121] S. Saremi, S. Mirjalili, A. Lewis, Grasshopper optimization algorithm: theory and application, *Adv. Eng. Software* 105 (Mar. 2017) 30–47.
- [122] A.A. Ewees, M.A. Elaziz, Z. Alameer, H. Ye, Z. Jianhua, Improving multilayer perceptron neural network using a chaotic grasshopper optimization algorithm to forecast iron ore price volatility, *Resour. Pol.* 65 (Mar. 2020) 101555.
- [123] Z. Elmi, M.O. Efe, Multi-objective grasshopper optimization algorithm for robot path planning in static environments, in: *Proc. IEEE Int. Conf. Ind. Technol. (ICIT)*, 24, Feb. 2018, p. 244.

- [124] A. Tharwat, E.H. Houssein, M.M. Ahmed, A.E. Hassanien, T. Gabel, MOGOA algorithm for constrained and unconstrained multiobjective optimization problems, *Int. J. Speech Technol.* 48 (8) (Aug. 2018) 2268–2283.
- [125] Timothy Oluwaseun Araoye, Evans Chinemezu Ashigwuike, Adeyinka Cornelius Adeyemi, Sochima Vincent Egoigwe, Nnaemeka Genesis Ajah, Emmanuel Eronu, Reduction and control of harmonic on three-phase squirrel cage induction motors with voltage source inverter (VSI) using ANN-grasshopper optimization shunt active filters (ANN-GOSAF), *Scientific African* 21 (2023).
- [126] sunpower, SunPower® SunPowerPerformance series | P17 ® performance series | p17-355-commercial-solar-panels. <https://us.sunpower.com/sites/default/files/media-library/data-sheets/ds-sunpower-p17-355-commercial-solar-panels.pdf>, 2016. (Accessed 2 May 2024).
- [127] T.O. Araoye, E.C. Ashigwuike, S.V. Egoigwe, F.U. Ilo, A.C. Adeyemi, R.S. Lawal, Modeling, simulation, and optimization of biogas-diesel hybrid microgrid renewable energy system for electrification in rural area, *IET Renew. Power Gener.* 15 (10) (Jul. 2021) 2302–2314, <https://doi.org/10.1049/rpg2.12164>.